Beyond the Matrix

The Algebraic Structure of Reality

Fundamental Fractal Geometric Field Theory (FFGFT)

From $T^4/\mathbb{Z}_3$ to the Particles of the Standard Model

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https://github.com/jpascher/T0-Time-Mass-Duality

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Chapter 1

Do We Live in a Matrix?

The question is old. Plato posed it with the allegory of the cave: people sit chained, see only shadows on the wall, and take these shadows for reality. Behind them stands the real light. Descartes formulated it as radical doubt: what if an evil spirit deceives us, what if everything we perceive is only a deceptively real illusion? And in 2003 Nick Bostrom gave it a modern, mathematically precise form: if sufficiently advanced civilizations can and want to run simulations of their ancestors, then we live with overwhelming probability in such a simulation.

The question is in vogue. Physicists like Max Tegmark argue that the universe is not only describable by mathematics — it *is* a mathematical structure. Computer scientists like Konrad Zuse and Stephen Wolfram suspect that nature computes on discrete rules, similar to a cellular automaton. Elon Musk said in an interview that the probability of not living in a simulation is one in billions.

For All Readers: *Imagine you are playing a video game so perfect that from inside you cannot tell whether it is a game. The physics within would be consistent, the mathematics would work, everything would follow rules — you would have no way to distinguish. That is Bostrom's argument.*

It is a serious question. And it is wrongly framed.

Not because the answer would be tedious. But because "simulation" is the wrong category. A simulation presupposes: someone computes. There is hardware outside. There is a simulator. The universe would then be an artifact — made, not being.

What if the universe is not a simulation *but* has a mathematical structure — deep, necessary, without external instance? What if the question is not: *who computes us*, but: *what is the structure that nature cannot be otherwise?*

That is a different question. And it has a different kind of answer.

Tegmark moves in this direction. His “Mathematical Universe Hypothesis” states: all mathematical structures exist — our universe is one of them. This is bold, but vague. It does not explain which structure. It makes no predictions. It is a philosophical position, not a physical theory.

For All Readers: *Tegmark says: all possible universes exist, and we happen to live in one of them. This is like saying: all possible melodies exist, and we happen to hear this one. It does not explain why this melody sounds the way it does.*

What this writing presents is a structure. Concrete, derivable, testable. No simulator, no external instance, no random universe from an ensemble. Instead: a geometric axiom — a compact four-dimensional torus with a threefold symmetry, mathematically $T^4/\mathbb{Z}_3$ — from which algebraically follows what must follow. No free parameters. No fits. No postulate about particles.

And that leads directly to the second question.

Chapter 2

Why Not Ten?

The Standard Model of particle physics is the most successful physical model ever formulated. It predicted the g-factor of the electron to twelve decimal places. It predicted the Higgs boson before it was found. It describes all known forces except gravity with terrifying accuracy.

And it has nineteen free parameters.

These nineteen numbers — particle masses, coupling strengths, mixing angles — are measured and inserted. The model does not explain why they have these values. It accepts them. It is as if one had a score that sounds perfect, but nobody knows by what principle it was composed.

For All Readers: *Imagine you find a clock. It runs precisely. You can predict where the hand will be in an hour. But you do not know why the clock is built the way it is — why twelve hours, why these gears, why this spring. The Standard Model is this clock. FFGFT asks: on what principle was it built?*

Among these parameters, one fact stands out particularly: there are three generations of fermions. Three times the same basic structure — electron, muon, tau; the associated neutrinos; three color states each for quarks. Nine fundamental families when counting color.

Why three? The Standard Model is silent. The LEP experiment at CERN showed in 1989 that there can be no fourth light neutrino generation — experimentally, through measurement of the Z-boson decay width. But "no fourth light generation" is not the same as "structurally impossible". There could be a heavy fourth generation. The experiment does not rule it out. It just does not find it.

For All Readers: *The LEP accelerator measured and said: we see no trace of a fourth generation here — at least not a light one. This is a finding, not a proof. It is like looking out the window and seeing no elephants — that does not rule out elephants in general, it just does not find them here.*

FFGFT gives a different answer. Not experimental — structural.

The starting point is $T^4/\mathbb{Z}_3$. This space has algebraic fixed points — points that are mapped onto themselves under the $\mathbb{Z}_3$ symmetry operation. These fixed points are not postulated. They are mathematically necessary: whoever folds T^4 with $\mathbb{Z}_3$ obtains exactly nine of them. Not eight. Not ten. Nine.

And these nine fixed points correspond — after an algebraic derivation carried out completely in the third part of this book — exactly to the nine fermion families of the Standard Model.

But this is only half the statement. The other half is more important:

A tenth family is algebraically forbidden.

Not experimentally not found. Not theoretically unlikely. Forbidden — in the same sense in which it is forbidden in chess to move the king two squares diagonally. Not because no one tried it. But because the rules of the system do not allow it.

For All Readers: *When you draw a triangle on a table, it has exactly three corners. Not because you stopped drawing. But because a triangle by definition has three corners — a fourth corner would make it a quadrilateral, not a larger triangle. This is how it is with the three generations in FFGFT: the algebraic structure of $T^4/\mathbb{Z}_3$ has exactly three eigenvalues. A fourth generation would not add a fourth corner — it would destroy the structure of the triangle.*

The algebraic core: The $\mathbb{Z}_3$-circulant — the mathematical representation of the threefold symmetry on T^4 — has exactly three eigenvalues. Not more. And these three eigenvalues, one of the central results of this theory, correspond to the three observed fermion generations (Doc. 282, 292, 348).

This is no fit. There is no free parameter set to three. The number three comes from the algebra, because $\mathbb{Z}_3$ has three elements and because the fixed point structure of the torus allows no other solutions.

The same holds for the particle charges. From the Legendre symbol modulo 13 — a number-theoretic function that naturally appears in the Galois field $GF(27)^*$ — it follows that electric charges can only take the values 0, ±1/3, ±2/3, ±1 (Doc. 346). Not ±1/2. Not ±1/4. Exactly the observed values — algebraically forced.

For All Readers: *The electric charge of a quark is 1/3 or 2/3 — never a half, never a quarter. That was always puzzling. Why thirds of all things? FFGFT gives an answer: because $GF(27)^*$ is a mathematical structure in which thirds are the only possible charge quanta. Not coincidence — algebraic necessity.*

A theory that says "there are nine" is impressive. A theory that says "there can be no others" — that is a different class of statement.

The Standard Model describes what is observed. FFGFT derives what must be observed — and why everything else is structurally excluded.

The difference is not academic. It concerns the question whether physics is description or understanding. Whether we measure parameters or discover principles. Whether the universe is the way it is because someone set it up that way — or because it cannot be otherwise.

For All Readers: *Imagine two maps. The first shows exactly where all the mountains are — precise, reliable, useful. The second not only shows that, but also explains why there must be mountains exactly there — the geology, the tectonics, the forces that lift the ground. Both maps are correct. But the second understands more.*

The Standard Model is the first map. FFGFT claims to be the second.

This claim is substantiated step by step in the following chapters — with complete derivations, with test scripts, with honest status markers that distinguish what is proven, what is sketched, and what is still open.

The universe is not a simulation. It is not a random selection from an ensemble of possible worlds. It is an algebraic structure — and this structure allows no other particles than those we know.

Chapter 3

What Mathematics Owes Nature

There is an old philosophical question that physicists like to overlook: When an equation describes nature — what exactly does it describe?

The obvious answer is: reality. But what does that mean more precisely? Three positions have emerged in the philosophy of science, and they lead to very different expectations of a physical theory.

The first position is called **Instrumentalism**: Mathematics is a tool. An equation is good when it delivers reliable predictions — nothing more, nothing less. Whether a deeper reality stands behind the equation is not a scientific question. The instrumentalist does not ask "what is the electron", he asks "what does the electron do".

The second position is called **Structuralism**: Reality *is* the mathematical structure. Not the equation describes the world — the equation *is* the world, in a precise ontological sense. Tegmark's Mathematical Universe Hypothesis is the most radical variant: all mathematical structures exist, our universe is one of them.

The third position — let us call it **geometric constitutionalism** — is more precise and more modest at the same time. It says: There is a geometric foundational structure. From it follow algebraic consequences — necessary, not postulated. These algebraic objects correspond to observable quantities. But whether these objects *are* what electrons and quarks *are* — that remains an open question. The theory claims structure, not essence.

For All Readers: *Imagine you discover that all rivers in a mountain range spring exactly where geology demands it — no exceptions, no coincidences. That is a deep insight. But it does not answer what water "is". The structural insight and the essence claim are different levels.*

FFGFT adopts the third position. This is not modesty from weakness — it is methodological precision. The status system of the corpus explicitly distinguishes four types of statements:

[SETZUNG] — A geometric axiom: $T^4/\mathbb{Z}_3$ as fundamental space. This is not derived, it is declared. Every theory needs a beginning, and this is the beginning of FFGFT.

[B] — Algebraically proven. What follows from the axiom with mathematical necessity. No free parameter, no fit, no discretion.

[K] — Derived from ξ, numerically tested. Quantitative predictions that have been compared with measured values.

[S] — Sketched. Plausible, structurally consistent, but the complete derivation is still missing. Honestly declared as such.

For All Readers: *A good carpenter says which joint he has glued and which he has only fitted. He does not say everything is glued when it is not. The status system makes FFGFT auditable — not because the theory is weak, but because it is honest.*

This distinction is unusual in theoretical physics. Most theories implicitly mix proven consequences with plausible assumptions. FFGFT makes the distinction explicit — because it wants to be testable, not just impressive.

The Ontological Boundary

Where does algebra end and where does being begin?

This question can be illustrated with the particle diagram. FFGFT derives algebraic objects from $GF(27)^*$ — classes f_1, f_2, f_3, f_4 with precise properties. These classes *correspond* to the lepton generations. They have the same charge structure, the same mass hierarchy, the same Majorana/Dirac separation.

But they *are* not electrons in space. They are algebraic invariants on a compact torus. Whether these invariants are ontologically *identical* to what a detector registers — that is a question FFGFT leaves open. Deliberately.

For All Readers: *A map shows exactly where a mountain is. The map is correct, complete, reliable. But the mountain on the map is not the mountain in reality — it is its representation. FFGFT says: The algebraic objects are the most precise representation of the particles we have.*

Whether they are identical to the mountain — we pose this question without answering it.

This sounds like retreat. It is the opposite.

Because the alternative — to claim that the algebra *is* reality — would be a leap that no physical theory can justify. Tegmark's structuralism is philosophically bold, but it is not physics. It makes no predictions that distinguish it from other positions.

FFGFT makes predictions. Concrete, falsifiable, numerically testable predictions. The Weinberg angle to 0.06 percent. The neutrino masses to under one percent. The Majorana/Dirac separation as a testable statement about neutrinoless double beta decay. That is physics.

Two Layers — Methodologically Separated

FFGFT works in two clearly separated levels (Doc. 241):

The first level is **ratio-based, parameter-free, exact**. Mass and coupling ratios without units, derived from the geometric structure. At this level there is no free parameter.

The second level sets **a single SI anchor** — such as the electron mass — and calculates all absolute values from it. Only here do units appear, only here is the connection to the measurement laboratory established.

This separation is not cosmetic. It makes visible what really follows from the theory and what is convention. $E = mc^2$ and $E = m$ are the same statement — once in SI units, once in natural units (Doc. 077). c is not a fundamental law, but a ratio — the ratio of length to time, fixed by the choice of convention.

For All Readers: *Celsius and Kelvin measure the same physical quantity — temperature. The zero point differs, the scale is identical. To say "0°C is a fundamental constant of nature" would be a category error. FFGFT shows that c is a similar category error — important as a convention, not fundamental as nature.*

What Mathematics Owes Nature — and What It Does Not

Mathematics owes nature descriptive accuracy. An equation must be correct if it claims to be correct. That is the duty.

Mathematics owes nature no essence statement. "What is an electron really" is not a mathematical question. Mathematics can say: The electron has this algebraic structure, these invariants, this charge. Nothing more.

FFGFT fulfills the duty — and does not cross the boundary. This is not a limitation. It is the condition for the theory to remain scientific.

The following chapters work within this framework. What is proven is marked [B]. What is derived is marked [K]. What is sketched is marked [S]. And what remains open is not concealed — it is named.

Chapter 4

The Torus

Geometry begins with shapes. And the simplest shape worthy of a physical fundamental theory is not a sphere and not a flat space — it is a torus.

A torus is, simply put, a surface like a donut. But this simplification conceals what is mathematically interesting. A torus arises not through a special curvature — it arises through **identification**. One takes a flat rectangle and declares: the right edge is identical to the left edge, and the top edge is identical to the bottom edge. The result is locally flat — no point of the torus "knows" that it lies on a torus — but globally closed.

For All Readers: *Imagine an old video game where the screen "wraps around" — when the player disappears at the right edge, it reappears on the left. The screen is a torus. It is locally flat — the figure moves straight ahead — but globally connected. Exactly this structure is the foundation of $T^4/\mathbb{Z}_3$.*

In physics, compactification on tori is an old idea. Kaluza and Klein proposed in the 1920s that a fifth dimension could be compact — so small that we cannot see it, but real enough to generate electromagnetism from pure geometry. The idea was correctly conceived, but too early.

FFGFT compactifies on T^4 — a four-dimensional torus. Four circles, identified with each other, locally flat, globally compact. And from this compactification follows the fundamental relation of the entire theory:

$\tilde{T} \cdot$ **m = 1**

This equation looks confusing at first. What does the intrinsic time $\tilde{T}$ have to do with the mass m?

The answer is geometric. On T^4, one of the four circles carries both the time structure and the mass structure. Mass is not an independent property attached to a particle — mass is the **inverse time scale** of a mode on the torus. A heavy particle has a short intrinsic time. A light particle has a long intrinsic time. Mass and time are not different — they are two sides of a relation.

For All Readers: *Imagine a guitar string. A thick, heavy string vibrates slowly — its frequency is low. A thin, light string vibrates fast. Mass and frequency stand in a fixed ratio. In FFGFT this is not just an analogy — it is the fundamental principle. The mass of a particle is its oscillation frequency on the torus, and conversely the intrinsic time is its inverse mass.*

This relation — $\tilde{T} \cdot$ m = 1 — is not postulated. It is a compactification relation: it follows from the fact that T^4 is compact and that one of its circles carries both mass and time. $\hbar$ and c are, from this standpoint, not fundamental constants of nature — they are SI conversion factors that arise when one measures length in meters and time in seconds instead of natural units.

The Euler Tonnetz as a Precursor

There is an astonishing analogy in music theory. The Euler Tonnetz — developed in 1739 by Leonhard Euler, expanded in the 19th century by music theorists — is a two-dimensional lattice in which tones are arranged by their frequency ratios. Horizontally lie pure fifths (factor 3/2), vertically pure thirds (factor 5/4). Every point in the lattice is a tone, every connection an interval.

What the Tonnetz shows: pitch is not an absolute property, it is a **ratio**. And the geometry of the lattice — which tones lie close together, which chords are possible, which modulations sound — follows from the structure of the lattice, not from the absolute frequency values.

For All Readers: *An octave is not "high" or "low" — it is a frequency ratio of 2:1. A fifth is 3:2. These ratios are the same in every key, whether you play in A or in B. The Tonnetz makes this ratio structure visible. FFGFT does the same for particle masses.*

In FFGFT, masses are not absolute values — they are ratios. The ratio of the muon mass to the electron mass is 206.77. This ratio follows from rational invariants on the torus. The absolute electron mass in kilograms is convention — it depends on how we define our measurement system. The ratio is structure.

The Tonnetz was a precursor. It showed that geometric structures force musical laws — not describe, force. FFGFT shows the same for physics: The geometric structure of $T^4/\mathbb{Z}_3$ forces the particle structure. Not describes — forces.

T^4 — Four Dimensions of the Torus

Why four? Not because spacetime has four dimensions — that would be a circular argument. But because the algebra demands it. The representation theory of the symmetry group, the classification of the fixed points, the structure of the Galois fields — all of this works consistently in four torus dimensions. Three would be too few for the complete particle structure. Five would be too many — they no longer force a unique solution.

This is not a proof of the uniqueness of T^4 — it is an observation. The theory sets T^4 as [SETZUNG]. What follows from it is [B] and [K].

In the next chapter, the threefold division is added — and with it the crucial switch that generates the nine fixed points and excludes all other solutions.

Chapter 5

The Threefold Division: $\mathbb{Z}_3$ and the Nine Fixed Points

T^4 alone is not enough. A four-dimensional torus is symmetric in every direction — it favors no structure, no number, no hierarchy. To obtain the particle structure from the geometry, a second step is needed: a symmetry operation that folds the torus onto itself.

This operation is called $\mathbb{Z}_3$. It is the group with three elements: the identity, a rotation by 120°, a rotation by 240°. When one identifies T^4 under this operation — that is: points connected by $\mathbb{Z}_3$ are declared the same point — the orbifold space $T^4/\mathbb{Z}_3$ arises.

For All Readers: *Imagine an equilateral triangle. If you rotate it by 120°, it looks the same as before. The three corners are "equivalent" under this rotation — one cannot distinguish them from outside. $\mathbb{Z}_3$ is exactly this symmetry: threefold rotation, three equivalent positions. $T^4/\mathbb{Z}_3$ is the torus where this equivalence is enforced.*

What arises from this? Almost everywhere $T^4/\mathbb{Z}_3$ is a perfectly normal space — smooth, without peculiarities. But at certain points something decisive happens: There lies a **fixed point**. A point that is mapped onto itself under the $\mathbb{Z}_3$ operation.

How many fixed points does $T^4/\mathbb{Z}_3$ have?

The answer is algebraically necessary: **nine**.

This is not the result of a calculation that could have been set up differently. It is a direct consequence of the geometry. A four-dimensional torus under a threefold identification has exactly nine invariant points — not eight, not ten, nine. [SETZUNG + B]

For All Readers: *When you fold a chessboard three times — once horizontally, once vertically, once diagonally — certain points arise that lie on themselves under all three folds. These points are completely determined by the geometry of the folding. Exactly so it is with the nine fixed points of $T^4/\mathbb{Z}_3$.*

What Happens at the Fixed Points

Fixed points are physically distinguished locations. At a regular point of the torus there is no preferred direction — the physics is homogeneous. At a fixed point the $\mathbb{Z}_3$ symmetry breaks locally — the point "knows" that it is special. And this local symmetry breaking generates localized modes: excitations that are bound to the fixed point.

These are the particles.

More precisely: The localized modes at the nine fixed points correspond to the nine fundamental fermion families of the Standard Model. Electron, muon, tau — three charged leptons. Three associated neutrinos. Three color states of quarks per generation, three generations — in total nine families. [B]

For All Readers: *Imagine a pond whose surface is smooth almost everywhere — but at nine specific spots there are vortices. These vortices do not arise by chance, they arise because the geometry of the pond forces vortices exactly there. The particles are these vortices. And the geometry of $T^4/\mathbb{Z}_3$ allows exactly nine of them — no more.*

The $\mathbb{Z}_3$-Circulant and the Three Generations

Within these nine fixed points there is a further structure. The $\mathbb{Z}_3$ operation acts not only globally — it also acts on the fiber over each fixed point. This fiber structure is a three-dimensional complex vector space $\mathbb{C}^3$, and the $\mathbb{Z}_3$ symmetry acts on it as a Hermitian circulant.

A Hermitian $\mathbb{Z}_3$-circulant is a special matrix — a 3×3 matrix that is invariant under cyclic permutation of its rows and columns. Such matrices have a special property: their eigenvectors are always exactly the three $\mathbb{Z}_3$ characters — the three Fourier modes of the threefold rotation. [B Doc. 282]

This means: The fiber structure over each fixed point automatically yields exactly three eigenstates. Three, no more, no less. And these three eigenstates are the **three generations**.

For All Readers: *A tuning fork vibrates at its fundamental frequency. But overtones are also possible — double frequency, triple, and so on. A $\mathbb{Z}_3$-circulant is like a tuning fork with exactly three allowed tones: the* *fundamental and two overtones. No fourth vibrational mode is possible — the mathematics of threefold symmetry allows exactly three. These are the three generations.*

The consequence is decisive: A **fourth generation is structurally excluded**. Not because the experiment does not find it. Not because it would be too heavy. But because the $\mathbb{Z}_3$-circulant has only three eigenvectors — algebraically, necessarily, without exception.

This is the difference from any experimental exclusion: The LEP experiment excludes a fourth *light* neutrino generation because it does not see it. FFGFT excludes a fourth generation of any kind because the algebra has no room for it.

The Frobenius Separation: Massive and Massless

Within $GF(27)^*$ — the Galois field with 27 elements, whose multiplicative group is $\mathbb{Z}_{26} \cong \mathbb{Z}_2 \times \mathbb{Z}_{13}$ — another algebraic operation acts: the Frobenius automorphism $x \mapsto x^3$.

This automorphism divides $GF(27)^*$ into two types: [B Doc. 339]

The **fixed points** +1, -1 — exactly two elements that are mapped onto themselves under $x \mapsto x^3$. These correspond to the massive sector: particles with m > 0. The fixed points are stable, localized, sluggish.

The **three-orbits** — eight groups of three elements each, which are run through cyclically under repeated application of $x \mapsto x^3$. These correspond to the massless sector: the eight gluons of the strong force.

For All Readers: *Imagine a clock whose hand keeps turning — those are the three-orbits, always in motion. And two positions where the hand stands still — those are the fixed points, that is the massive sector. The separation between massive and massless is not an imposed distinction — it follows directly from the algebra of $GF(27)^*$.*

The photon — massless particle of the electromagnetic field — corresponds to the fixed field $GF(3)$ of the Frobenius, the U(1) sector. This too is not postulated, it follows from the algebra structure. [K]

Four Classes — Dirac and Majorana

The four irreducible cubic polynomials over $GF(3)$ — f_1, f_2, f_3, f_4 — define the four fundamental orbit types in $GF(27)^*$. Each orbit has three elements. Three elements — three generations. [B Doc. 348]

The sector pairing k $\mapsto -k$ (particle/antiparticle) and the Majorana criterion k $\mapsto$ k^{-1} mod 13 divide these four classes into two layers:

- f_1, f_2: **Majorana layer** — neutrinos ν_1 and ν_2, their own

- f_3, f_4: **Dirac layer** — charged leptons (e, μ, τ) and

And ν_3 — the third neutrino — lies in the Dirac layer. It is **not a Majorana particle**. This is a falsifiable prediction: The neutrinoless double beta decay, which is only possible for Majorana neutrinos, should not occur for ν_3. [B Doc. 343]

For All Readers: *Some particles are their own mirror image — Majorana particles. Others have a true opposite — Dirac particles. Which neutrinos are which type has not yet been decided experimentally. FFGFT gives a clear answer: ν_1 and ν_2 are Majorana, ν_3 is Dirac. This is testable — and if it is wrong, part of the theory is wrong.*

This is the character of FFGFT: It makes predictions that could be wrong. Not because the theory is uncertain, but because that is the condition for science.

The nine fixed points. The three generations. The Majorana/Dirac separation. Everything follows from $T^4/\mathbb{Z}_3$ — from the geometry that one sets once, and the algebra that follows necessarily from it.

Chapter 6

$\tilde{T} \cdot m = 1$: Time and Mass as Dual

There is a question that physicists have avoided for a century: What is time?

Relativity theory made it a fourth dimension — it stretches, it passes more slowly at high speed, it curves in gravitational fields. But what it *is* remains open. Is time a container in which things happen? Is it a property of matter? Is it fundamental, or does it emerge from something deeper?

Quantum mechanics gave a surprising answer — not intentionally, but as a structural finding. Wolfgang Pauli proved in 1933 that for a Hamiltonian with a spectrum bounded below there can be no self-adjoint time operator. The consequence: Time is not an observable in quantum mechanics. One cannot measure *when* a system is in a state — one must prescribe time from outside, as a parameter.

Time is kept out of the Hilbert space. It stands beside it.

For All Readers: *In quantum mechanics there are operators for position, momentum, energy, spin — everything one can measure. But no operator for time. Time is the framework, not the content. It runs in the background, like an external timekeeper — but it is not part of the physics happening in the foreground. That is not a technical detail. It is a structural problem.*

FFGFT gives a different answer. Not through a new quantum mechanics — but through a geometric insight that places time and mass in a single relation:

$\tilde{T} \cdot$ **m = 1**

This equation is short. What it says is radical.

What the Equation Means

$\tilde{T}$ is the intrinsic time of a mode — not the external, classical time that runs in the background, but a time scale that belongs to the mode itself. And m is the mass of this mode. The product is one.

This means: Mass and intrinsic time are not two independent properties. They are inversely coupled. A heavy particle has a short intrinsic time. A light particle has a long intrinsic time. The heaviest known elementary particle — the top quark, about 186 times heavier than a proton — has an intrinsic time scale of about 10^{-25} seconds. The electron, the lightest charged particle, has an intrinsic time scale about 200,000 times longer.

For All Readers: *Imagine a clock that runs faster the heavier it is. A kilogram-clock runs faster than a gram-clock. That sounds strange — but that is exactly what $\tilde{T} \cdot m = 1$ says. Not for ordinary clocks, but for particles: their intrinsic time scale is the reciprocal of their mass. Heavy and fast, light and slow.*

This is not an analogy. It is a compactification relation on T^4. One of the four circles of the torus carries both the mass structure and the time structure of a mode. Mass is the winding frequency on this circle. Time is the reciprocal. Both are the same geometric quantity — seen from two directions. [SETZUNG + K, Doc. 077]

E = mc² = E = m

From $\tilde{T} \cdot$ m = 1 follows immediately an insight that sounds familiar and unfamiliar: Einstein's famous equation E = mc^2 is in natural units identical to E = m. Not approximately — exactly. [B Doc. 077]

In natural units c = 1 holds — not because the speed of light is uninteresting, but because it is a ratio: length over time. If one measures length and time in the same units, this ratio is one. Then E = mc^2 becomes E = m $\cdot$ 1^2 = m.

The factor c^2 in Einstein's equation is not a law of nature — it is a unit conversion factor. It arises because we measure length in meters and time in seconds, instead of in a consistent unit. [B]

For All Readers: *Temperature can be measured in Celsius or in Kelvin. The physics does not change. Only the numerical value changes, and the formulas that use the convention carry conversion factors. c^2 in E = mc^2 is such a conversion factor — it carries the information that we measure length and time in different units. In natural units it disappears.*

This sounds like a marginal simplification. It is more than that. Because it means: Mass *is* energy, without reservation, without conversion factor, without the detour through an “equivalence”. They are the same object — in natural units even numerically identical.

And if mass is energy, and energy determines the time scale — then mass, energy and time are a single thing, seen from three directions.

The Problem of Superposition

$\tilde{T} \cdot$ m = 1 has an unexpected consequence for the interpretation of quantum mechanics — specifically for the superposition principle.

In standard quantum mechanics, a system can be in a superposition of two states: $\psi = c_1|\psi_1\rangle + c_2|\psi_2\rangle$. This state is timeless — it has no intrinsic time structure. Pauli had ensured that: time stands outside.

In FFGFT, each mode has its own intrinsic time scale $\tilde{T}_i$ = 1/m_i. A superposition of two modes with different masses is a superposition of two modes with *different intrinsic time scales*. This is structurally inconsistent: One mode “lives” on the scale 1/m_1, the other on the scale 1/m_2. A cross-modal superposition generates incompatible time scales. [B Doc. 334]

For All Readers: *Imagine two musicians playing together — one in 3/4 time, one in 4/4 time, and neither may give way. The result is no stable ensemble — it is a structural tension. In FFGFT, the superposition of two particles with different masses is similar: Their intrinsic time scales are different, and this is not a minor detail — it is a geometric fact about the torus.*

This does not solve the measurement problem of quantum mechanics — it reframes it. The question is no longer ”why does the superposition collapse upon measurement”, but ”why is a cross-modal superposition possible in FFGFT at all, and how long is it stable”. This is an open bridge — honestly declared as [S].

Two Types of Unrolling

$\tilde{T} \cdot$ m = 1 describes a compact circle S^1 — the mass mode on the torus. When an observer measures, they find a real time axis $\mathbb{R}$, not a compact circle. What happens at the transition?

FFGFT distinguishes two processes, both of which can be called “unrolling”, but are fundamentally different: [B Doc. 333]

The first — **Type I, the measurement act** — is lossy. The observer chooses the mass circle as the time axis and discards the winding number. The discrete spectrum of the compact circle disappears. What remains is the continuous time axis of classical physics. This loss is real and irreversible.

The second — **Type III, the coordinate change** — is lossless. The function space on the compact circle $L^2(S^1)$ is isomorphic to the space of periodic functions on $\mathbb{R}$. This is a representation change — no information is lost, the Fourier basis remains preserved, the pullback is exact.

For All Readers: *The difference is like between a map that one folds (Type I — one loses overview, can no longer see how the edges connect), and a map that one transfers to a different coordinate system (Type III — Mercator instead of Lambert, but the same information, only differently represented).*

This distinction is not academic. It decides which physical statements are transferable and which require an interpretation of the measurement act. What holds on the compact torus — winding numbers, discrete modes, topological invariants — holds after Type III transfer also in $\mathbb{R}$. What is only accessible after the measurement act (Type I) has undergone a loss.

Time as an Emergent Quantity

The deepest consequence of $\tilde{T} \cdot m = 1$ is philosophical: Time is not a fundamental quantity. It emerges from the mass structure of the torus.

In standard physics, time stands as an external parameter at the beginning — the equations describe how things behave *in time*. In FFGFT it is the opposite: The torus T^4 is the fundamental structure. Time as a continuous variable arises only when an observer performs the measurement act — when they choose the mass circle as the time axis and discard the winding number.

This means: There is no universal time. Each mode has its own time scale $\tilde{T}_i = 1/m_i$. The coordinated time in which we all live "simultaneously" is a macroscopic approximation — the result of the fact that macroscopic objects consist of billions of atoms whose individual time scales disappear in the statistical average and leave an effective, common time scale. [K]

For All Readers: *In an orchestra, hundreds of instruments play. No single instrument sets the tempo — the common tempo emerges from the interplay. In FFGFT, time is like that. No fundamental clock ticks in the background. The time we experience emerges from the mass structure of the world in which we move.*

$\tilde{T} \cdot$ m = 1 is a single equation. It contains no free parameters. And it changes what mass, energy and time mean — not as metaphor, but as a mathematically precise, algebraically necessary consequence of the geometry of T^4.

Chapter 7

Finite Fields: When 2 + 2 Is Not 4

The mathematics behind the particle structure of FFGFT is not the mathematics of real numbers. It is an older, rarer, in physics hardly used mathematics: the theory of **finite fields** — Galois fields, named after Évariste Galois, who developed them in the 19th century before dying at age 20 in a duel.

A finite field is a mathematical system in which one can add, subtract, multiply and divide — as with ordinary numbers — but the system contains only finitely many elements. This sounds paradoxical: division is after all unbounded, how is that supposed to work with finitely many numbers?

The answer is: calculation modulo a prime number.

For All Readers: *Imagine a clock. It has twelve hours — the numbers 0 to 11. When you calculate 10 + 5, you get not 15 but 3 (because 15 = 12 + 3, and the clock starts again at 0 after 12). That is calculation modulo 12. On a prime clock — say modulo 3, with the numbers 0, 1, 2 — even division works: 1 ÷ 2 = 2, because 2 × 2 = 4 = 3 + 1 ≡ 1 (mod 3). Every number except 0 has a reciprocal. That is* $GF(3)$ *— the smallest non-trivial Galois field.*

$GF(3)$ **— The Beginning**

$GF(3)$ is the Galois field with three elements: 0, 1, 2. Addition and multiplication are modulo 3. The special feature: In $GF(3)$, $-1 = 2$, because $1 + 2 = 3 \equiv 0 \pmod 3$. And $2 \times 2 = 4 \equiv 1 \pmod 3$ — so 2 is its own reciprocal.

Why is $GF(3)$ physically relevant? Because the threefold symmetry $\mathbb{Z}_3$ of the torus has exactly three elements — and because the algebraic representation of this symmetry builds on $GF(3)$. The $\mathbb{Z}_3$ charge operator, which classifies color and charge, acts in $GF(3)$. [B Doc. 341]

For All Readers: *The three colors of quarks — red, green, blue — are described in quantum chromodynamics by the group SU(3). In FFGFT this threefold structure is already in $GF(3)$: The three elements 0, 1, 2 encode the three color states. Not as metaphor — as algebraic fact.*

$GF(9)$ **— The Next Level**

$GF(9)$ is the Galois field with nine elements. It cannot be constructed as a simple residue class modulo 9 — because 9 = 3^2 is not prime, and in $\mathbb{Z}/9\mathbb{Z}$ there is no unique division (3 × 3 = 9 ≡ 0, so 3 has no reciprocal). Instead, $GF(9)$ is constructed as an extension of $GF(3)$: One adds an element i with the property $i^2 = -1$ (which in $GF(3)$ means the equation $i^2 = 2$, since $-1 = 2$ mod 3). This is analogous to the construction of complex numbers, where one adds i with $i^2 = -1$ — only that here everything takes place in $GF(3)$. [B]

For All Readers: *Complex numbers arise by adding to the real numbers a new number i satisfying $i^2 = -1$. $GF(9)$ arises analogously: one adds to $GF(3)$ a new number i satisfying $i^2 = 2$ (the $GF(3)$ equivalent of -1). The result is a field with exactly nine elements: 0, 1, 2, i, i+1, i+2, 2i, 2i+1, 2i+2.*

The multiplicative group $GF(9)^*$ — that is, $GF(9)$ without zero — has eight elements. These eight elements correspond to the eight gluons of the strong nuclear force. This is no coincidence and no fit — it is a direct algebraic consequence of the Frobenius separation in $GF(27)^*$. [B Doc. 339]

$GF(27)$ **— Where the Particles Live**

$GF(27)$ is the Galois field with 27 elements — the third in the series $GF(3) \subset GF(9) \subset GF(27)$. The construction proceeds as before: One adds to $GF(3)$ an element that satisfies an irreducible cubic equation. Cubic — because 27 = 3^3.

The multiplicative group $GF(27)^*$ has 26 elements. 26 = 2 × 13. And this decomposition is the key to the entire particle structure. [B Doc. 339]

The $\mathbb{Z}_2$ component encodes particle/antiparticle — parity. The $\mathbb{Z}_{13}$ component decomposes into four three-orbits under the Frobenius automorphism $x \mapsto x^3$. Each orbit has three elements — three generations. Four orbits — four particle classes: neutrinos, charged leptons, up quarks, down quarks. Everything from the algebra. [B]

For All Readers: *$GF(27)^*$ is like a crystal with 26 atoms. The Frobenius automorphism is a rotation operation — it rotates each orbit through its three positions. When one analyzes the crystal under this rotation, one finds exactly the structure of the known elementary particles. Not because one constructed it in — but because the algebraic structure of the crystal forces it.*

The Inclusion $GF(3) \subset GF(9) \subset GF(27)$

This nesting of Galois fields is not random — it corresponds to a nesting of physical structures:

$GF(3)$ — color charge. Three values 0, 1, 2, three colors, the $\mathbb{Z}_3$ symmetry of the torus.

$GF(9)$ — the electroweak structure. Eight elements in $GF(9)^*$, eight gluons. The Weinberg angle $\sin^2\theta_W = 0.2308$ follows from the $GF(9)$ topology. [K]

$GF(27)$ — the complete fermion structure. 26 elements, four orbit classes, three generations per class, Majorana/Dirac separation, charge quantization.

Each level of the inclusion corresponds to a level of the physical structure hierarchy. This is the deepest finding of FFGFT: The hierarchy of forces and particles of the Standard Model is the hierarchy of Galois fields $GF(3^n)$ for n = 1, 2, 3.

The Legendre Symbol and Charge Quantization

Why do quarks carry electric charges of ±1/3 and ±2/3? Why not ±1/2? The Standard Model has never answered this question — it sets the charges as measured facts.

$GF(27)^*$ gives an answer. The four three-orbits O_1, O_2, O_3, O_4 divide into two classes according to the **Legendre symbol modulo 13**: [B Doc. 346]

O_1 = 1, 3, 9 and O_3 = 4, 10, 12 consist of quadratic residues modulo 13. The Legendre symbol (k/13) = +1. These orbits carry electric charge Q = 0: neutrinos.

O_2 = 2, 5, 6 and O_4 = 7, 8, 11 consist of quadratic non-residues modulo 13. The Legendre symbol (k/13) = -1. These orbits carry electric charge Q ≠ 0: charged leptons and quarks.

And the color charge — encoded by k mod 3 in $GF(3)$ — determines whether a charged particle is a lepton (color singlet) or a quark (color triplet). The combination of both properties completely classifies all fermions of the Standard Model. [B]

For All Readers: *A quadratic residue modulo 13 is a number that can be written as the square of another number modulo 13. For example, 3 is a quadratic residue modulo 13, because $4^2 = 16 \equiv 3 \pmod{13}$. And 2 is not a quadratic residue modulo 13. This number-theoretic property — quadratic residue or not — decides whether a particle is electrically charged or not. This is the deepest known explanation of charge quantization.*

The Harmonic Primes: 2, 3, 5, 7, 11, 13

The Galois fields $GF(3^k)$ have multiplicative groups of order $3^k - 1$. A prime p appears in this group order exactly when k is a multiple of $\text{ord}_p(3)$ — the smallest positive number k for which $3^k \equiv 1$ (mod p). [B Doc. 342]

The first four physically relevant primes after 3 appear at: - k = 3: p = 13, because $3^3 - 1 = 26 = 2 \times 13$ - k = 4: p = 5, because $3^4 - 1 = 80 = 16 \times 5$ - k = 5: p = 11, because $3^5 - 1 = 242 = 2 \times 11^2$ - k = 6: p = 7, because $3^6 - 1 = 728 = 8 \times 7 \times 13$

These primes 2, 3, 5, 7, 11, 13 are exactly those in which all nine Yukawa coefficients of the fermions lie. This is no coincidence — it is algebraically forced by the structure of the Galois field hierarchy. [B Doc. 358]

For All Readers: *In music there are harmonic tones — overtones that lie at integer multiples of the fundamental frequency. Not all frequencies sound harmonious together, only certain ones. In FFGFT there are harmonic primes — primes that appear in the Galois field orders that fit the torus geometry. And exactly these primes build the particle masses. Algebra and music follow the same deeper principle: structure resonance.*

p = 13 is particularly distinguished: It is the only prime worldwide with $\text{ord}_p(3) = 3$. This means: Exactly p = 13 appears at k = 3 — in $GF(27)$, the field of the fermion structure. No other prime has this property. [B]

The finite fields are not the language that physicists usually speak. They are the language in which nature has written the particle structure.

Chapter 8

The Galois Lattice of Particle Masses

If the algebra is correct, it must calculate. Not approximately — precisely.

This is the moment when a physical theory decides: Does it remain philosophical or does it become physics. FFGFT becomes physics.

An important external comparison came from Douglas Matzke (IPI, September 2026), who developed the GALG framework — an independent algebraic description of physical structures via Clifford algebras. Matzke's systematic search in GALG G(3) (6561 attempts) found no element of order 26 — exactly the algebraic prohibition that FFGFT predicts from the $GF(27)^*$ structure. The agreement is no coincidence: G$(3) \cong M_2(GF(9))^2$ contains no order-26 elements algebraically necessarily. Order 26 first lives in G$(6) \cong M_8(GF(9))$ — and exactly there sits the FFGFT mass layer for leptons. [B Doc. 341]

The Mass Formula

The mass of a fermion i in FFGFT is (Doc. 006, 317): [K]

$m_i = r_i \cdot \xi^{p_i} \cdot v \cdot K_{\text{frak}}$

Here v = 246 GeV is the electroweak vacuum expectation value — the only element that couples to the SI unit system. ξ = 4/30000 is the only free parameter of the entire theory — and as the following chapter shows, it too is algebraically forced, not a measured value. K_frak = 74/75 is the fractal-recursive correction — derived from the Hausdorff dimension of the torus, no fit.

The quantities r_i and p_i are the actually interesting ones: they are **winding numbers on the torus** — integer invariants that describe the topological class of the fermion mode. For the leptons: [K]

Particle	n_θ	n_φ	$r_i = n_\varphi^2/n_\theta$	p_i
Electron	3	2	4/3	3/2
Muon	5	4	16/5	1
Tau	9	5	25/9	2/3

These numbers are not fitted. They follow from the topology of the fermion figure on T^4: n_θ is the winding number in the θ direction, n_φ in the φ direction. The recursion n_θ,g = n_θ,g−1 + n_φ,g−1 connects the generations — 3, 5=3+2, 9=5+4. This is a Fibonacci-like structure that follows from the Galois field hierarchy.

For All Readers: *Imagine a rope wound around a cylinder. The more often it is wound, the heavier the particle. The electron is wound three times in one direction, twice in another. The muon five times and four times. The winding numbers are topological numbers — they cannot change continuously, they jump from one integer to the next. That is why the particles have discrete masses.*

In the Ratio: K_frak Disappears

In the ratio of two lepton masses, K_frak cancels exactly. [B] This is methodologically important: What follows from the ratio is independent of the fractal correction — a cleaner prediction.

$m_\mu/m_e = (r_\mu/r_e)\cdot\xi^{(p_\mu-p_e)} = (12/5)\cdot\xi^{-1/2} = (12/5)\cdot\sqrt{7500} = 120\sqrt{3} \approx 207{,}846$

The experimental value (PDG): 206.768. Deviation: +0.52

This is the bare prediction — without K_frak, without QED corrections, without fits. 0.52parameter ξ.

For All Readers: *An archer aims without wind correction and hits to 0.52"almost right" — that is a result that demands explanation. Either the archer was lucky, or he knows the principle. FFGFT knows the principle: The 0.52measurements.*

The Key Identity: (m_μ/m_e)^2 = 43200

The square of the mass ratio removes the irrational $\sqrt{3}$ and yields an exact integer identity — the same number from two completely different derivations: [K Doc. 338]

(r_μ/r_e)^2/ξ = $|GF(9)^*|^2\cdot 5^2\cdot|GF(27)|$ = $8^2\cdot 25\cdot 27$ = **43200**

On the left is the FFGFT side: winding numbers of the torus, divided by ξ. On the right is the Galois side: group orders of the

Galois fields, multiplied. Both sides yield the same number — exact, without approximation, without coincidence.

The factors on the Galois side have physical meaning: - 8 = $|GF(9)^*|$: the eight gluons of the strong force - 27 = $|GF(27)|$: the field of the third fermion generation - 5: the smallest new prime in $GF(3^4)^*$ = first prime beyond the $GF(27)$ horizon

For All Readers: *Imagine you calculate the same value twice — once by measuring distances in a triangle, once by solving an equation — and obtain exactly the same number. That proves not only that both methods are correct. It proves that the triangle and the equation have the same structure. Exactly this is what 43200 shows: The torus geometry and the Galois algebra have the same deep structure.*

ξ Is Not a Free Constant

From the identity $(m_\mu/m_e)^2$_FFGFT = 43200_Galois, ξ follows necessarily: [K]

ξ = (r_$\mu/r_e)^2$ / 43200 = (144/25) / 43200 = 4/30000

ξ is not a free constant — it follows algebraically from the Galois group orders and the winding numbers of the torus. The only parameter of the theory is not a parameter. It is a result.

This is the core: FFGFT has no free parameter in the proper sense. What looks like a free parameter — ξ — is algebraically forced. What remains is a single choice of convention: the coupling to SI units via the electroweak vacuum expectation value v. And this too is structurally determined — by $\tilde{T}\cdot$ m = 1 as a compactification relation.

The Yukawa Coefficients in the Galois Lattice

All nine Yukawa coefficients of the fermions lie in the prime lattice 2, 3, 5, 7, 11, 13. This is the central finding of Doc. 358: [B]

Electron: r_e = 4/3 = $2^2/3$ ✓ Muon: r_μ = 16/5 = $2^4/5$ ✓ Tau: r_τ = 25/9 = $5^2/3^2$ ✓

Earlier it seemed that the strange quark (coefficient 13, k=3) and the top quark (coefficient 7, k=6) fell out of the lattice. Doc. 358 shows: They do not fall out. They lie at k=3 and k=6 — exactly the levels at which p=13 and p=7 enter the Galois field hierarchy. No exception. No coincidence. Algebraic necessity.

For All Readers: *A scale has specific notes — not all possible frequencies, but selected ones. When an instrument can only play these notes, that is not a restriction of its freedom — it is its nature. The particle masses play on a Galois scale: only the primes 2, 3, 5, 7, 11, 13 are allowed, because* *only they appear in the Galois field hierarchy that belongs to the $T^4/\mathbb{Z}_3$ geometry.*

The Fine-Structure Constant from ξ

For completeness: The fine-structure constant $\alpha \approx 1/137$ also follows from ξ. The precise Galois derivation yields [K Doc. 338]:

$1/\alpha = 3700/27 = 137.037...$

This is from pure Galois group orders — without ξ, without v, without m_e as input. The experimental value (PDG): 137.036. Deviation: 7.6 ppm.

For All Readers: *The fine-structure constant 1/137 is one of the most mysterious numbers in physics. Feynman said every theorist should write it on the wall and ask why it is exactly this. FFGFT gives an answer: because 3700/27 = 137.037 — and 3700 and 27 are Galois group orders that follow necessarily from the $T^4/\mathbb{Z}_3$ geometry.*

What Remains Open

The bare prediction deviates from experiment by 0.5–1.6not measurement uncertainty — the PDG errors are below 0.0001fractal-recursive correction of higher order already contained in the measured values. Whether this correction can be fully derived — that is the open bridge R113. Honestly declared as [S].

Physics is not the art of talking away corrections. It is the art of understanding them.

Chapter 9

Three Generations: Algebraic Prohibition of the Fourth

In the previous chapter we saw that the lepton winding numbers follow a recursion: 3, 5=3+2, 9=5+4. Three levels. And then?

There is no fourth level.

This is not an observation. It is not an experimental result. It is an algebraic theorem — with proof.

The $\mathbb{Z}_3$-Circulant and Its Three Eigenvalues

The mass operator in FFGFT is a Hermitian $\mathbb{Z}_3$-circulant on the $\mathbb{C}^3$ fiber over each torus fixed point. This is a 3×3 matrix of the form:

[B Doc. 282]

M = [[a, b, c], [c, a, b], [b, c, a]]

with complex entries a, b, c, where a is real (Hermiticity). This matrix is invariant under cyclic permutation of rows — exactly the property that $\mathbb{Z}_3$ symmetry demands.

Hermitian $\mathbb{Z}_3$-circulants have a known property: Their eigenvectors are always exactly the three $\mathbb{Z}_3$ characters

ω^0 = (1, 1, 1)/$\sqrt{3}$ ω^1 = (1, ω, ω^2)/$\sqrt{3}$ ω^2 = (1, ω^2, ω^4)/$\sqrt{3}$

where $\omega = e^{2\pi i/3}$. These three eigenvectors are completely determined by the $\mathbb{Z}_3$ symmetry — independent of the entries a, b, c. This means: The generation structure follows from the symmetry, not from the specific mass parameters.

And the number of eigenvectors of a 3×3 matrix is three. No more.

For All Readers: *A 3×3 matrix has exactly three eigenvectors — that is linear algebra, school material. A room with three walls has three corners. No interior designer can add a fourth corner without rebuilding the room. Just so: The $\mathbb{Z}_3$-circulant has three eigenvalues — the three generations. A fourth would destroy the symmetry, not extend it.*

Three Generations from Frobenius Orbits

On the Galois side, the same statement comes differently packaged. The four primitive orbits O_1, O_2, O_3, O_4 in $GF(27)^*$ each have three elements — because the Frobenius automorphism $x \mapsto x^3$ has order 3 on $\mathbb{Z}_{13}$. The three elements of each orbit correspond to the three generations. [B Doc. 348]

This is not an analogy — it is the same algebraic statement in two languages. The $\mathbb{Z}_3$-circulant eigenvector and the Frobenius orbit are the same mathematical object, described with the tools of Hilbert space theory and Galois field theory respectively.

In both descriptions: three. Necessary.

Comparison with the LEP Argument

The LEP experiment at CERN measured the Z-boson decay width in 1989 with great precision. This width depends on how many channels the Z-boson can decay into — and among these are the neutrino-antineutrino pairs. The result: The measured width corresponds exactly to three light neutrino generations.

This is an excellent experiment. But it only rules out light neutrinos — those whose mass is significantly smaller than half the Z-boson mass (≈ 45 GeV). A fourth generation with heavy neutrinos (mass > 45 GeV) would have escaped the LEP experiment. And quarks of a fourth generation would have had other production channels anyway.

The LEP argument is: We see no fourth light generation. The FFGFT argument is: There is no fourth generation of any kind.

Argument	Type	Scope
LEP	Experimental	Light neutrinos (m < 45 GeV)
FFGFT	Algebraic	All fermion generations

For All Readers: *LEP looks through a telescope and says: we see no fourth planet in this section of the sky. FFGFT calculates the laws of* *gravity and says: in this solar system no further stable orbits are possible. Both statements can be true simultaneously — but the second is deeper.*

The Coupling Matrix: Permutation and 1/$\sqrt{2}$

Not only the number of generations follows from the Galois structure — also the coupling between the sectors. The trace bilinear form on $GF(27)$ over $GF(3)$ — Tr_$GF(27)$/$GF(3)$(ab) — yields the coupling matrix between the four orbit classes. [B Doc. 348]

The result is remarkable:

$f_3 \times f_4$ (charged leptons × quarks): The coupling matrix is a permutation matrix. This means: The coupling between charged leptons and quarks has the maximum possible symmetry — each lepton couples to exactly one quark, with equal strength.

$f_1 \times f_3$ (neutrinos × charged leptons): The entries are 1/$\sqrt{2}$. This is the maximum mixing angle — 45°. The large mixing in the neutrino sector, which experiment confirms, is algebraically predicted.

The mixing angles themselves — CKM matrix for quarks, PMNS matrix for neutrinos — are not yet fully derived from this. Their structure is [B], their exact values are open [S]. This is an honest statement about the state of the theory.

For All Readers: *In physics there are matrices that describe how strongly particles of different generations couple to each other. These matrices have certain entries — numbers between 0 and 1. FFGFT says: Some of these entries are 1 (maximum coupling), some are 1/$\sqrt{2}$ (45° mixing), some are 0 (no direct coupling). This follows from the Galois algebra, not from experiment.*

Electrically Neutral and Colored — Algebraically Forbidden

A final consequence of the Galois structure, experimentally confirmed but never explained: There is no electrically neutral colored fermion.

In the Standard Model this is an observed fact — but not an explained one. There could be such particles in another world.

In FFGFT it is algebraically forbidden. The combination QR/NQR (electric charge) and N mod 3 (color charge) from $GF(27)^*$ yields four possible classes: neutral+colorless (neutrinos), charged+colorless (charged leptons), neutral+colored — this class does not exist in $GF(27)^*$, because QR always coincides with color singlet —, and charged+colored (quarks). [B Doc. 346]

The empty class is no coincidence. It is an algebraic consequence of the Galois structure of $GF(27)^*$. An electrically neutral colored fermion would have no correspondence in the fixed point structure of $T^4/\mathbb{Z}_3$.

It cannot exist — because the algebra has no room for it.

Chapter 10

Gluons, Color, and Confinement

In the previous chapter we saw how the nine fixed points of $T^4/\mathbb{Z}_3$ generate the nine fermion families. But the Standard Model has more than fermions. It has force particles — bosons that mediate the interactions between fermions. And the strongest of these interactions is the strong nuclear force — mediated by eight gluons.

Where does the number eight come from? The Standard Model sets it as a property of the gauge group SU(3). FFGFT derives it.

Gluons as Galois Objects

In $GF(27)^*$ with its 26 elements, the Frobenius automorphism $x \mapsto x^3$ acts. We have seen that it decomposes the group into fixed points +1, -1 and three-orbits. The fixed points are the massive sector — the fermions. What are the orbits?

The four three-orbits in $\mathbb{Z}_{13}$ — $\{1, 3, 9\}$, $\{2, 5, 6\}$, $\{4, 10, 12\}$, $\{7, 8, 11\}$ — can be combined with the $\mathbb{Z}_2$ parity from $\mathbb{Z}_{26} \cong \mathbb{Z}_2 \times \mathbb{Z}_{13}$. This gives: 4 orbits × 2 parities = **8 objects.** [B Doc. 339]

These are the eight gluons.

And the three elements of each orbit — the three Frobenius cycles within an orbit — correspond to the **three colors** red, green, blue. Not postulated. Algebraically forced by the three-period structure of the Frobenius.

For All Readers: *Imagine a pattern of eight groups of three positions each — 24 elements total plus two fixed points makes 26. That is $GF(27)^*$ completely partitioned. The eight groups are the gluons, the three positions the colors, the two fixed points the massive matter. No free parameter. All from a single group structure.*

The photon comes from another corner of the same structure: the **fixed field** $GF(3)$ of the Frobenius on $GF(27)$. The projection $GF(27) \to GF(3)$ yields $GF(3)^*$ = 1, 2 $\cong \mathbb{Z}_2$ — a particle that is its own antiparticle. The photon. [B Doc. 339]

Structure element	Physical correspondence
Fixed points +1,−1 in $GF(27)^*$	Massive sector (fermions)
4 orbits × 2 parities = 8	Eight gluons
3 elements per orbit	Three colors (R, G, B)
Fixed field $GF(3)^*$ = 1,2	Photon, U(1) sector

Why Quarks Cannot Exist Alone

This is the deepest question of the strong nuclear force: Why has no one ever seen a single quark? Quarks always come in groups — mesons (quark + antiquark) or baryons (three quarks, one per color). This is called **confinement**.

The Standard Model describes confinement numerically — lattice calculations show that the energy between two quarks grows linearly with distance, so that separating them would cost infinite energy. But it does not explain why.

FFGFT gives an algebraic answer: **Confinement is $\mathbb{Z}_3$ triality selection.** [B Doc. 321/325]

Color charge in FFGFT is the $\mathbb{Z}_3$ charge operator N mod 3 — an element of $GF(3)$ with values 0, 1, 2. An observable object — a particle that emerges from the torus into observable space — must be **$\mathbb{Z}_3$-invariant**. That is: its total triality must be zero, N mod 3 = 0.

A single quark has N mod 3 = 1 or 2 — it is not $\mathbb{Z}_3$-invariant. It cannot emerge from the torus as an isolated object. Three quarks with colors R, G, B have together N mod 3 = 1+1+1 = 3 ≡ 0 — invariant, observable. A quark-antiquark pair has N mod 3 = 1+2 = 3 ≡ 0 — also invariant. [B]

For All Readers: *Imagine a doorman rule: Only groups whose sum is a multiple of three may enter. A single guest with ticket "1" cannot enter. Three guests with "1", "1", "1" — sum 3, so multiple of 3 — can enter. Two guests with "1" and "2" — sum 3 — can enter. That is $\mathbb{Z}_3$ triality. That is confinement.*

This rule is not imposed from outside — it follows from the geometry of the torus. Only $\mathbb{Z}_3$-invariant modes can survive the Type II transition (the geometric projection from T^4 into three-dimensional observable space). All others are not observed because they cannot exist algebraically.

The Proton: Three Quarks, $\mathbb{Z}_3$-Invariant

A proton consists of two up quarks and one down quark (uud). Each carries a color — R, G, B — so that the total triality is 0.

In $GF(27)^*$, up and down quarks belong to the NQR sector (quadratic non-residues mod 13). Their electric charges +2/3 and $-1/3$ follow from the Legendre symbol [B Doc. 346] — as shown in Chapter 7.

The proton has charge 2×(+2/3) + 1×(−1/3) = +1. This is no coincidence — it is an algebraic consequence of charge quantization from the Legendre symbol.

The proton mass itself — 938 MeV — does not follow directly from the Galois structure, but from lattice QCD: the strong interaction between the three quarks and the gluon field generates the largest part of the mass. This is [S] for FFGFT — an area where the theory is still working. Honestly declared.

For All Readers: *The proton weighs 938 MeV. The electron weighs 0.511 MeV. The proton is therefore 1836 times heavier. But the three quarks in the proton weigh only about 10 MeV together — almost nothing. The rest is generated by the gluon field, which holds the three quarks together and converts enormous energy into mass (E = mc²). The gluon field itself is an algebraic consequence of $GF(27)^*$ — and its number, eight, is algebraically forced. What the exact mass gives is a calculation, not a derivation from geometry — still.*

The Complete Galois Structure of the Standard Model

With the gluons, the picture is complete. $GF(27)^*$ classifies all particles of the Standard Model:

Fermions: QR sector (leptons, neutrinos) and NQR sector (quarks) — in total 9 families × 3 colors × 2 chiralities [B Doc. 346–349]

Gluons: 8, from 4 orbits × 2 parities [B Doc. 339]

Photon: from the fixed field $GF(3)^*$ [B Doc. 339]

Electroweak bosons W±, Z^0: from the Galois group Gal($GF(9)/GF(3)$) = $\mathbb{Z}_2$ [K Doc. 336]

Confinement: $\mathbb{Z}_3$ triality selection, no free parameter [B Doc. 325]

The only particle still open in FFGFT: the Higgs boson. Experimentally confirmed [X] — but the FFGFT derivation of the Higgs mechanism is not yet complete. In Doc. 354 an approach is sketched: $\tilde{T} \cdot m = 1$ could replace or generate the Higgs mechanism. That is [S] — open, precisely named.

Chapter 11

Antiparticles, Spin, and Chirality: States, Not Doublings

When counting Standard Model particles, one quickly encounters confusion. There are twelve fermions — but then antiparticles too: the positron for the electron, the antiquark for the quark. Does that make 24? And then spin-up and spin-down for each. Does that make 48?

The answer is: no. And the explanation lies in the algebra.

What an Antiparticle Is

An antiparticle is not a different particle. It is the same algebraic object — with reversed charge. The positron has the same mass as the electron, the same spin, the same interaction strength. Only the electric charge is opposite: $+1$ instead of -1.

In $GF(27)^*$ this difference corresponds to the $\mathbb{Z}_2$ parity — the factor $\mathbb{Z}_2$ in the decomposition $\mathbb{Z}_{26} \cong \mathbb{Z}_2 \times \mathbb{Z}_{13}$. Particle and antiparticle are the two $\mathbb{Z}_2$ elements of the same orbit. [[B] Doc. 339]

They are not two separate objects. They are one algebraic object with two states — like a coin with two sides, not two different coins.

For All Readers: *Imagine a clock. The hands can turn clockwise or counterclockwise. That is not a second mechanism — it is the same mechanism* *in two possible states. Particle and antiparticle are the same algebraic object, once with charge $+q$, once with $-q$. The $\mathbb{Z}_2$ parity in $GF(27)^*$ is exactly this reversal state.*

What Spin Is

Spin is even more fundamental. It is not a spinning ball — that historical picture failed in 1925 because it would require unphysical rotation speeds. Spin is an algebraic property following from the structure of space.

In the Clifford algebra Cl(6) — the algebraic description of the six-dimensional internal space of FFGFT — every state has a grade r determining how it transforms under rotations. Spin-1/2 particles correspond to grade-1 objects (vectors) in Clifford space.

The key: a spin-1/2 particle requires a **rotation by 720°** to return to itself — not 360°. This is not a paradox. It follows from the algebraic structure:

$$\gamma_5 \cdot X = (-1)^r \cdot X$$

For odd grades r: $\gamma_5 X = -X$ — this is **left-handed**. For even grades: $+X$ — **right-handed**. [[B] Doc. 349]

Spin-up and spin-down are also not two different particles. They are two projections of the same algebraic object onto different spatial directions.

For All Readers: *A shadow on a wall can be long or short, depending on how the light falls. The object is the same — the shadow is a projection. Spin-up and spin-down are projections of the same spinor onto different measurement directions. Not two objects — one object, two measurement states.*

What Chirality Is

The weak nuclear force behaves strangely: it acts only on left-handed particles and right-handed antiparticles. This is one of nature's most peculiar asymmetries — **parity violation**, discovered in 1956.

In FFGFT this asymmetry follows from the Cl(6) grade parity — from the same γ_5 operator that defines spin. The **Su sector** (even Cl(6) grades) is right-handed. The **Sd sector** (odd Cl(6) grades) is left-handed. And the weak force $SU(2)_L$ couples only to the Sd sector — algebraically forced, not postulated. [[B] Doc. 349]

The right-handed neutrino ν_R is structurally absent: in the Sd sector there is no room for it. This is consistent with the prediction from the neutrinos chapter: ν_3 is Dirac, but ν_R as an independent degree of freedom does not exist in $GF(27)^*$. [[B]/[S] Doc. 349]

The Complete Count

What are the "12 particles" really?

Property	Algebraic origin	Number of states
Particle families	9 fixed points of $T^4/\mathbb{Z}_3$	9
Color	N mod 3 in $GF(3)$	$\times 3$ for quarks
Particle/antiparticle	$\mathbb{Z}_2$ parity in $GF(27)^*$	$\times 2$
Chirality	Cl(6) grade parity	$\times 2$
Spin projection	Measurement direction	2 states

None of this is an independent doubling. Everything is states of the same algebraic object — different projections, parities, grades within a single structure.

The "12 fermions" of the Standard Model are in FFGFT: 3 leptons (QR sector, colorless) + 3×3 quarks (NQR sector, 3 colors) = 12 *family-color combinations*. Antiparticles, spin, and chirality are not additional objects — they are algebraic states within the same Galois structure.

Chapter 12

The Higgs Boson: Particle or Vacuum Artifact?

In July 2012 CERN announced the discovery of the Higgs boson — a particle with mass 125 GeV, predicted by the Standard Model since 1964. A triumph. But a question was left unanswered in the celebration:

What is the Higgs boson, really?

What the Standard Model Says

In the Standard Model there is a Higgs field Φ permeating all of space. This field has a **vacuum expectation value** $\langle\Phi\rangle = v \approx 246$ GeV — it is non-zero in the vacuum. Particle masses arise from interaction with this background: $m = y \cdot v$, where y is the Yukawa coupling.

The Higgs boson H^0 is the quantum oscillation of this field around its equilibrium — the excitation observed at the LHC in 2012.

Why does the vacuum have this non-vanishing value? Because the Higgs potential $V(\Phi) = \mu^2|\Phi|^2 + \lambda|\Phi|^4$ with $\mu^2 < 0$ has its minimum not at $\Phi = 0$ but at $|\Phi| = v$. Why is $\mu^2 < 0$? The Standard Model answers: *it is set that way. That is how it is.*

For All Readers: *Imagine a valley. The Standard Model says: the universe sits in the valley because the valley is there. Why is the valley there? Because the potential has this shape. Why does it have this shape? No answer. FFGFT asks differently: what geometry forces this valley?*

What FFGFT Says

In FFGFT, $\tilde{T} \cdot m = 1$ is not a postulate about fields — it is a compactification relation on T^4. And this relation has a direct consequence: **a vacuum with $m = 0$ is algebraically excluded.** [[B] Doc. 354]

If $m = 0$, then $\tilde{T} \to \infty$ — an infinitely long intrinsic time scale. That is physically unrealizable on a compact torus. The torus *forces* mass in the vacuum. What the Standard Model sets through $\mu^2 < 0$ follows in FFGFT from geometry.

The Higgs field Φ and the time field $\tilde{T}$ are **inverse to each other**:

[[B] Doc. 354]

$$\tilde{T}(x,t) = \frac{1}{m(x,t)} = \frac{1}{y \cdot (v + h(x,t))}$$

The time field is the inverse Higgs field. The Higgs mechanism is the *flat projection* of the compact T^4/Z_3 topology into the language of quantum field theory.

The spontaneous symmetry breaking $SU(2)_L \times U(1)_Y \to U(1)_{\text{EM}}$ follows from three Galois building blocks: [[B] Doc. 355]

- $\tilde{T} \cdot m = 1$ forces $\langle \Phi \rangle \neq 0$
- Z_3 Galois structure forces $Q_{\text{vac}} = 0$ (neutral vacuum)
- $Q = 0$ selects $U(1)_{\text{EM}}$ as residual group

Does the Higgs Boson Exist?

This question has a precise answer in FFGFT — in two parts.

Part 1: The Higgs field exists. The non-vanishing vacuum expectation value is forced, not postulated. [B]

Part 2: Whether the Higgs boson H^0 is an independent excitation or an artifact of the flat description — that is open. [[S] Doc. 354]

The particle observed at 125 GeV is real [X]. The question is what it *is*: A true excitation of a fundamental field? Or the representation of a torus topology in the language of flat quantum field theory — useful as a description, but not fundamental?

For All Readers: *Sound is real — we hear it. But sound is not fundamental: it is a pressure wave in air. In a theory without air there is no sound. The Higgs boson may be similar: real as an observation, but not fundamental — a wave in a vacuum field that is itself a projection artifact. FFGFT says: possibly. Not yet proved.*

The most honest statement FFGFT can make: the Higgs mechanism follows from geometry [B]. The Higgs boson as a particle is experimentally established [X]. Whether it is fundamental or a projection — open [S].

Chapter 13

Neutrinos: Majorana or Dirac?

Neutrinos are the most mysterious particles of the Standard Model. They have mass — we know that since the neutrino oscillation experiments of the late 1990s. But how large this mass is, we only know approximately. And whether neutrinos are their own antiparticles — Majorana particles — or whether they have a true opposite — Dirac particles — has not yet been decided experimentally.

FFGFT gives a clear answer. Not speculative — algebraically proven.

Majorana or Dirac: What Is the Difference?

A Dirac particle has a true antiparticle — a particle with opposite charge and otherwise identical properties. Electron and positron are Dirac particles. A Majorana particle is its own antiparticle — there is no distinction between particle and antiparticle. For charged particles this is impossible: An electron cannot be its own antiparticle because the positron carries opposite charge. But neutrinos are electrically neutral — for them Majorana is possible in principle.

For All Readers: *Imagine a picture that in mirror image looks the same as the original. That is a Majorana particle: its mirror image is identical to itself. Electron and positron, by contrast, are like a picture and its negative — clearly different. For charged particles this is forced. For neutrinos it is an open question — until FFGFT.*

The Algebraic Answer

In Chapter 9 we saw that the four orbit classes f_1, f_2, f_3, f_4 divide into two layers — through the sector pairing k $\mapsto -k$ (particle/antiparticle) and the Majorana criterion k $\mapsto$ k^{-1} mod 13.

The Majorana criterion is algebraically precise: An orbit is Majorana if it is mapped onto itself under the inversion k $\mapsto$ k^{-1} mod 13.

Orbit 3 = 4, 10, 12: 4^{-1} = 10 mod 13, 10^{-1} = 4 mod 13, 12^{-1} = 12 mod 13. This orbit is self-inverse. It carries Majorana character. [B Doc. 340]

Orbit 4 = 7, 8, 11: 7^{-1} = 2 mod 13 — lies in Orbit 2, not in Orbit 4. This orbit is not self-inverse. It carries Dirac character.

The consequence for the neutrinos: - ν_1 (Orbit 1) and ν_2 (Orbit 3): **Majorana** — self-inverse, own antiparticle - ν_3 (Orbit 4): **Dirac** — not self-inverse, has true antiparticle

This is a falsifiable prediction. Neutrinoless double beta decay — a process only possible for Majorana neutrinos — should occur for ν_1 and ν_2, not for ν_3. The KamLAND-Zen experiment is searching for it. FFGFT says: $m_{\beta\beta} \leq$ 14.4 meV — only from ν_1 and ν_2 contributing, ν_3 not. [B]

For All Readers: *There is an experiment searching for neutrinoless double beta decay — a process in which two neutrinos annihilate each other. This is only possible if neutrinos are their own antiparticles. FFGFT says: This occurs for ν_1 and ν_2, but not for ν_3. If the experiment finds a different result, part of the theory is wrong.*

Neutrino Masses from the Galois Structure

Not only the Majorana/Dirac classification — also the neutrino masses themselves follow from the $GF(27)^*$ structure. [K Doc. 340]

The base neutrino mass is $m_\nu = \xi^2/2 \cdot m_e \approx$ 4.54 meV. This is still a setting — the factor $\xi^2/2$ is plausible from the torus geometry, but not yet fully derived ([S]).

What then follows is algebraically precise. The maximum element of the fourth Frobenius orbit is 11 — because:

11 = $|\mathbb{Z}_{13}|$ – |Frobenius fixed points in $\mathbb{Z}_{13}|$ – 1 = 13 – 1 – 1 = 11

This number determines the mass hierarchy:

Neutrino	Galois origin	Mass
ν_1	Orbit 1 1,3,9	$m_\nu \approx$ 4.54 meV
ν_2	Orbit 3 4,10,12 self-inverse	$\sqrt{14/3} \cdot m_\nu \approx$ 9.81 meV
ν_3	Orbit 4 7,8,11	$11 \cdot m_\nu \approx$ 49.96 meV

For All Readers: *The number 11 is not arbitrarily chosen. It follows necessarily from the structure of $GF(27)^*$: 13 elements in $\mathbb{Z}_{13}$, minus 2 fixed points, minus 1 — makes 10, and 10+1=11 as the maximum element of the heaviest neutrino orbit. Three neutrinos, three masses, all from the same algebraic source.*

The Mass Differences: Comparison with Experiment

The neutrino oscillation experiments do not measure absolute masses — they measure differences of the mass squares. Two differences are known:

The atmospheric mass difference: $\Delta m^2_{\text{atm}} = \mathrm{m}^2(\nu_3) - \mathrm{m}^2(\nu_1) = (11^2 - 1) \cdot \mathrm{m}_\nu^2 = 120 \cdot \mathrm{m}_\nu^2$

Calculated: 2.476×10^{-3} eV^2 Measured (PDG): 2.500×10^{-3} eV^2 **Deviation: 1.0%** [K]

The factor $120 = 11^2 - 1$ is also the order of the icosahedral symmetry group $A_5 \times \mathbb{Z}_2$ — another algebraic connection not yet fully understood ([S]).

The solar mass difference: $\Delta m^2_{\text{sol}} = \mathrm{m}^2(\nu_2) - \mathrm{m}^2(\nu_1) = (11/3) \cdot \mathrm{m}_\nu^2$

The factor 11/3 follows from: $(|\mathbb{Z}_{13}| - 2)$ / Frobenius order = $(13 - 2) / 3 = 11/3$

Calculated: 7.565×10^{-5} eV^2 Measured (PDG): 7.530×10^{-5} eV^2 **Deviation: 0.46%** [K]

Both predictions to under one percent — from the Galois structure of $GF(27)^*$, without fits, without free parameters for the mass ratios.

For All Readers: *Neutrino physicists took decades to measure these mass differences. FFGFT derives them from the algebraic structure of $GF(27)^*$ — from a theory developed for the geometry of the universe at the smallest scale, not for neutrinos in particular. That the same lattice describes both is no coincidence.*

Inverted Hierarchy Excluded

Another prediction that is experimentally testable: FFGFT rules out the inverted neutrino mass hierarchy. In the inverted hierarchy, ν_3 would be the lightest neutrino, ν_1 and ν_2 the heavier ones. In FFGFT, ν_3 is the heaviest — that is the normal hierarchy.

The reason: In the inverted hierarchy, ν_1 and ν_2 would have Majorana character and be heavier — then $m_{\beta\beta}$ (the effective Majorana mass in neutrinoless double beta decay) would be $\approx$ 49 meV. KamLAND-Zen has set the limit $m_{\beta\beta} <$ 36 meV. The inverted hierarchy is therefore, if ν_1 and ν_2 are Majorana particles, experimentally excluded — consistent with the FFGFT prediction. [B]

Sum of Neutrino Masses

Cosmology sets a limit on the sum of all neutrino masses: $\Sigma m_i <$ 120 meV (Planck 2018). FFGFT yields:

$\Sigma m_i = (1 + \sqrt{14/3} + 11) \cdot m_\nu \approx 64.3$ meV [K]

Consistent with the cosmological bound — and close enough that future experiments (KATRIN, Euclid) can test the prediction.

For All Readers: *The universe has set a limit on the total mass of all neutrinos — too heavy, and structure formation in the early universe would have been disrupted. FFGFT says: The neutrino masses sum to 64 meV. This lies comfortably below the limit, but close enough that the next generation of experiments could see it.*

Neutrinos are the particles that interact most weakly with the world. Millions pass through every square centimeter of our bodies per second without leaving a trace. And yet their algebraic structure — Majorana or Dirac, which mass, which hierarchy — carries the clearest fingerprints of the Galois geometry of $T^4/\mathbb{Z}_3$.

Chapter 14

One Parameter, All Constants

The physics of the 20th century has left a list of constants of nature that look like arbitrarily chosen digits. Speed of light c. Planck's constant $\hbar$. Gravitational constant G. Fine-structure constant α. Boltzmann constant k_B. Each has a precisely measured value. None has an explanation for why this particular value.

Feynman put it succinctly: The fine-structure constant 1/137 is the greatest mystery of physics. A dimensionless number that seemingly comes from nowhere and yet determines all of chemistry and atomic physics. Every theorist should write it on the wall and ask why it is exactly this.

FFGFT writes it on the wall — and gives an answer.

First Step: Most Constants Are Conventions

Before ξ comes into play, it is worth seeing how many of the apparent constants of nature are in fact unit conventions.

In natural units — $\hbar$ = c = G = k_B = 1 — all mechanical, thermodynamic and gravitational quantities collapse to a single dimension: energy. [B, Doc. 241, 261]

Quantity	Dimension in natural units
Mass, energy, temperature, frequency	[E]
Length, time, wavelength	$[E]^{-1}$
Speed	dimensionless
Electric charge	dimensionless

$\hbar$, c, G, k_B are not constants of nature — they are exchange rates between different scale choices. Meter, second, kilogram, Kelvin are human conventions. Nature knows only one dimension.

For All Readers: *The exchange rate between Euro and Dollar is not a property of the universe — it depends on how much a Euro is worth, which is again convention. Just so, c, $\hbar$, G are exchange rates between different scale choices (second, meter, kilogram). In a consistent unit they are all equal to one — and disappear from the equations.*

What remains when all unit conventions are removed are **dimensionless numbers** — pure ratios, independent of scale choices. The fine-structure constant $\alpha \approx 1/137$ is one of them. The ratio of proton mass to electron mass $m_p/m_e \approx 1836$ is another. These numbers are true constants of nature — they have the same value in every unit system.

ξ — The One Parameter

FFGFT derives all dimensionless constants of nature from a single parameter: $\xi = 4/30000$.

As shown in Chapter 8, ξ itself is algebraically forced — it follows from the Galois group orders and the winding numbers of the torus. The only parameter of the theory is thus not a free parameter. It is a consequence of the geometry.

The derivation chain — complete, without fits, in five steps [K Doc. 180]:

1. The T0 Lagrangian with $\tilde{T} \cdot m = 1$ enforces a self-consistency condition
2. This condition plus the loop structure in four spacetime dimensions
3. The electron-proton-muon system determines ξ uniquely
4. From ξ follows G — the gravitational constant
5. From ξ follows $L_0 = \xi \cdot \ell_P$ — the minimal length scale

For All Readers: *Imagine a clockmaker's workshop where all gears mesh. You know the size of a single gear — all others follow necessarily. ξ is this one gear. The gravitational constant, the fine-structure constant, the minimal length scale — all follow from it, without additional inputs.*

The Fine-Structure Constant

Two independent derivations from ξ:

Derivation 1 — via the geometric formula [K Doc. 011/043]: $\alpha = \xi \cdot (E_0/1\ \text{MeV})^2$

where $E_0 = \sqrt{(m_e \cdot m_\mu)} \approx 7.348$ MeV is the geometric mean energy of the leptons. Result: α^{-1} = 137.06 (deviation 0.02%).

Derivation 2 — from pure Galois group orders [K Doc. 338]: $1/\alpha$ = 3700/27

Here 3700 = $|GF(9)^*|^2 \cdot 5^2 \cdot 2$ and 27 = $|GF(27)|$. No ξ, no v, no m_e as input. Result: α^{-1} = 137.037 (7.6 ppm).

Both derivations from the same geometric source. Both to under 0.03accuracy. This is no coincidence.

For All Readers: *Feynman said 1/137 comes from nowhere. FFGFT shows: It comes from 3700/27. And 3700 and 27 are group orders that follow algebraically from $T^4/\mathbb{Z}_3$. This is not an answer that answers all questions — but it is an answer that goes deeper than "that's just how it is".*

The Weinberg Angle

$\sin^2\theta_W$ = 0.2308 — the Weinberg angle of the electroweak theory, one of the most important coupling strengths of the Standard Model.

From the $GF(9)$ topology follows [K Doc. 336]: $\sin^2\theta_W = 1/4 \cdot (1 - 1/|GF(9)^*|^2) = 1/4 \cdot (1 - 1/64) \approx 0.2305$

The experimental value (PDG at M_Z): 0.23122. Deviation: 0.06%.

For All Readers: *The Weinberg angle determines how strongly the weak nuclear force and electromagnetism are mixed with each other — one of the fundamental properties of nature. From the algebra of $GF(9)^*$ with its 8 elements, this angle follows to 0.06structure.*

The Gravitational Constant

G — the weakest of the four forces, and the most difficult to integrate into quantum physics. FFGFT derives G from ξ [K Doc. 180]:

$G = \xi \cdot \ell_P^2/(\hbar/m_e \cdot c)$

In natural units: $G = \xi \cdot (L_0/\ell_P)^2$ — the gravitational constant is a geometric ratio, determined by ξ and the minimal length scale $L_0 = \xi \cdot \ell_P$.

This is a deep statement: Gravity is not one of four independent forces — it follows from the same geometric structure as all other constants. No free parameter, no fit.

The Cosmological Constant

The most famous fine-tuning problem in physics: The measured cosmological constant Λ is about 123 orders of magnitude smaller than the quantum field theoretic prediction. This is the largest discrepancy between theory and experiment in the history of physics.

FFGFT transforms this question. The 122 decades decompose algebraically into two contributions [K Doc. 312]:

122 decades = 77.5 (from ξ^{20}) + 44.8 (from $(\ell_P/\bar{\lambda}_e)^2$) − 0.4 (order 1)

From a fine-tuning question becomes a structure question: Why exactly level 10 of the ξ ladder? This is open — declared as open bridge P20, status [S]. But the transformation of the question is already a result.

For All Readers: *The fine-tuning problem of the cosmological constant is like the following puzzle: You know that the answer is approximately 10^{-123}. Why not 10^{-120} or 10^{-126}? FFGFT says: The answer decomposes into two algebraic contributions, each following from the Galois structure. Why exactly this decomposition — that is still open. But the question is now a different, more precise one.*

What ξ Does Not Explain

Honesty first: ξ explains the dimensionless ratios — the structural constants of nature. It does not explain the absolute energy scale — why the electron mass is exactly 0.511 MeV and not 0.511 GeV. This scale is set by the SI anchor v = 246 GeV — the electroweak vacuum expectation value, the only point where FFGFT couples to the measurement laboratory. [Doc. 241]

But this one anchor then determines all absolute masses — electron, muon, tau, all quarks, all bosons. One anchor, everything else follows.

For All Readers: *A map at scale 1:100,000 describes all ratios correctly — this mountain is three times as high as that one, this river is twice as long as that one. To know how high the mountain is in meters, one needs a reference point: a known distance on the map, measured in reality. That is v = 246 GeV in FFGFT. One measurement point — and all absolute values follow from the ratios.*

One parameter. All constants. And the parameter itself is no free choice — it is algebraically forced by the geometry of $T^4/\mathbb{Z}_3$.

Chapter 15

Leptons and Their Residuals

In Chapter 8 we saw: The bare prediction of the lepton masses lies 0.5–1.6beside experiment. This gap is small. But it is real, and it is systematic — the bare prediction lies consistently slightly too high.

This is not a sign of failure. It is information.

What the Residual Is — and What It Is Not

The residuals ε_i are defined as (Doc. 352):

$\varepsilon_i = (m_i^{\text{PDG}} - m_i^{\text{FFGFT}})/(m_i^{\text{PDG}} \cdot \xi)$

These are the deviations in units of ξ. For the three leptons: [K]

Ratio	FFGFT bare	PDG	Residual in ξ-units
m_μ/m_e	207.846	206.768	$+39.1 \cdot \xi$
m_τ/m_μ	16.992	16.817	$+77.9 \cdot \xi$
m_τ/m_e	3531.6	3477.2	$+117.4 \cdot \xi$

The residuals are not random — they are systematically increasing: 39, 78, 117 — threefold, in equal steps. [K]

What is the cause?

QED Self-Energy: Excluded

The most obvious explanation would be: QED radiation corrections. In quantum electrodynamics, particles receive a mass correction through interaction with virtual photons — the QED self-energy.

FFGFT tests this rigorously. The QED self-energy has the form δm/m $\propto \alpha \cdot$ ln(Λ/m), where Λ is a cutoff parameter. If this were the cause of the residuals, the residuals would have to grow logarithmically with mass — that is, $\delta\varepsilon/\varepsilon \propto$ ln(m_i).

The ratio of the observed residuals is 77.9/39.1 ≈ 1.99 ± 0.013. The logarithmic ratio ln(m_τ)/ln(m_μ) would be something else — and the ratio 1.99 deviates from it by **112 standard deviations**.

QED self-energy as the cause is excluded at 112σ. [K Doc. 352]

For All Readers: *112 standard deviations means: The probability that QED self-energy is the cause is so small that it would not have occurred a single time in the entire history of the universe. This is not a statistical exclusion — this is a structural exclusion.*

What Remains

Ten possible causes were systematically tested. Nine are excluded. The only surviving candidate: The residuals grow proportionally to the difference of the winding numbers Δn_θ between adjacent generations — that is, proportionally to n_φ of the lighter generation. The ratio 2 agrees to 0.64 standard deviations.

This is [S] — speculative, a hint. No complete derivation. Bridge R113 is open.

But the open bridge is precisely formulated — and that is worth more than a vague claim. A falsifiable prediction from the hint: m_τ = 1776.78 MeV. The PDG value: 1776.86 ± 0.12 MeV. The prediction lies 0.08 MeV below the measured value — within measurement uncertainty, but at the edge.

For All Readers: *A detective has excluded nine suspects and has one left. He has no proof yet — but he knows where to look. FFGFT has excluded nine explanations and has one left. The open bridge is not a defeat — it is the map for the next expedition.*

The Koide Scalar

A related observation: The Japanese physicist Yoshio Koide discovered in 1982 empirically that the three lepton masses satisfy a remarkable relation:

Q = $(m_e + m_\mu + m_\tau) / (\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2$ = 2/3

This relation is accurate to 0.04has tried to explain it. Koide did not know why. It looked like a coincidence.

FFGFT calculates from the Galois structure: Q_FFGFT = 0.6677 — a deviation of 0.04

This is not a proof of the Koide relation — Q_FFGFT is not exactly 2/3. But it is also no coincidence: The calculated value lies close to 2/3, because the Galois structure of $GF(27)^*$ has a broken symmetry that generates almost-but-not-quite 2/3. What generates the 0.04that is another open bridge. [S]

***For All Readers:** The Koide relation is like a melody that sounds beautiful on almost all instruments — but nobody knows why. FFGFT plays the melody almost perfectly: Q = 0.6677 instead of 2/3 = 0.6667. The tiny deviation of 0.04error — it is a signal pointing to a deeper structure not yet fully understood.*

Methodological Balance

What Chapter 12 shows is as important as the results themselves: how FFGFT deals with deviations.

Not ignore. Not redefine. Not dismiss as measurement error. But: precisely quantify, systematically test, honestly declare what is explained and what is not. Test nine explanations and exclude nine — that is more work than claiming one explanation and moving on.

The open bridges are not weaknesses of the theory. They are its next steps.

Chapter 16

The Zeta Function of the Torus

There is a connection between the particle structure of the universe and the deepest open problem in mathematics. It is not speculative — it is algebraically precise. And it shows how far the Galois structure of $T^4/\mathbb{Z}_3$ reaches.

The connection is called the zeta function.

What the Riemann Zeta Is

The Riemann zeta function $\zeta(s)$ is one of the most important functions in mathematics. Defined as the sum of all negative s-th powers of the natural numbers:

$\zeta(s) = 1 + 1/2^s + 1/3^s + 1/4^s + \ldots$

It has a deep property: It can be written as a product over all primes — the Euler product:

$$\zeta(s) = \prod_p \frac{1}{1 - p^{-s}}$$

Every prime contributes a factor. The distribution of primes is completely encoded in the zero structure of $\zeta(s)$. The Riemann hypothesis — unproven for 165 years — claims that all non-trivial zeros of $\zeta(s)$ lie on the line Re(s) = 1/2.

For All Readers: *Imagine you want to understand how the primes are distributed — why some gaps are large and others small. The Riemann zeta function is the tool for that: every prime leaves a trace in it, and the zeros of the function encode the entire distribution. That is the deepest unsolved problem in mathematics.*

What the Torus Has to Do with Primes

The spectral zeta function of a torus — the zeta function of the Laplace operator on T^4 — is in one dimension identical to the Riemann zeta. The spectrum of the Laplace operator on a circle is exactly the set of natural numbers. The zeta function of this spectrum is $\zeta(s)$.

When one folds the torus by $\mathbb{Z}_3$ — when one forms $T^4/\mathbb{Z}_3$ — the zeta function changes. The $\mathbb{Z}_3$ projection selects only $\mathbb{Z}_3$-invariant modes: those for which n ≡ 0 (mod 3). All other modes — n ≡ 1 or n ≡ 2 (mod 3) — form the twist sectors.

The result is a Dirichlet L-function: [B Doc. 343]

$$\zeta_{T^4/\mathbb{Z}_3}(s) = (1 - 3^{-s})\,\zeta(s) = L(s, \chi_0)$$

The factor $(1 - 3^{-s})$ is precise: It removes the contributions of the prime p=3 from the Euler product. Because p=3 is the geometric foundation of the orbifold — it stands outside the prime hierarchy, it defines the structure, it does not contribute as an Euler factor.

For All Readers: *When you free a song from everything the fundamental tone contains — all tones that are multiples of the fundamental — a peculiar, tension-filled structure remains. The orbifold zeta is the Riemann zeta freed from everything that contains p=3. What remains is the spectral signature of $\mathbb{Z}_3$ symmetry breaking.*

The Hierarchical Euler Product

The L-function L(s, χ_0) has an Euler product over all primes except 3. But these primes are not equal — they are ordered by their entry depth k = $\mathrm{ord}_p(3)$: [B Doc. 343]

k	Prime	Euler factor	Harmonic	FFGFT
1	2	4/3 = 1.333	Octave	$\mathbb{Z}_2$ fixed point, massive sector
3	13	170/169 ≈ 1.006	Tredecime	$GF(27)^*$, leptons
4	5	25/24 ≈ 1.042	Major third	Factor in $(m_\mu/m_e)^2$
5	11	122/121 ≈ 1.008	Undecime	Neutrinos
6	7	49/48 ≈ 1.021	Septime	Top quark
≥ 7	≥ 41	$\approx 1+p^{-2}$	—	suppressed

The first five primes 2, 5, 7, 11, 13 are exactly the harmonic primes from Chapter 7. Their Euler factors correspond to musical intervals — octave, third, septime, undecime, tredecime.

This is no coincidence and no wordplay. The spectral structure of the torus and the harmonic intervals of music follow the same mathematical principle: the Euler product by entry depth.

For All Readers: *In the 18th century, Euler developed the Tonnetz — a geometric representation of musical intervals as prime ratios. Every fifth is a factor 3/2, every third 5/4. The Euler factors of the orbifold zeta are the same primes in the same order — but now as contributions to the spectral zeta function of the torus. Euler and FFGFT describe the same structure.*

The New Zeros: Signature of $\mathbb{Z}_3$ Symmetry

The factor $(1 - 3^{-s})$ in the orbifold zeta brings new zeros that do not exist in the ordinary Riemann zeta: [B Doc. 343]

$$3^{-s} = 1 \iff s = 2\pi ik/\ln 3,\ k \in \mathbb{Z} \setminus \{0\}$$

These zeros lie on the imaginary axis Re(s) = 0 — not on the critical line Re(s) = 1/2 of the Riemann hypothesis. They arise exclusively through the $\mathbb{Z}_3$ orbifold projection. They are the algebraic signature of sector separation — the trace of the $\mathbb{Z}_3$ symmetry in the zeta function.

Important: These new zeros do not resonate with the known Riemann zeros. A resonance test of the first eight Riemann zeros against the orbifold zeros shows no significant agreement — the deviations lie between 0.15 and 0.47. [B]

FFGFT touches the Riemann hypothesis — it dances with it. But it does not solve it, and it does not claim to solve it.

For All Readers: *The Riemann hypothesis concerns the zeros of the full Riemann zeta, which encodes all primes. The orbifold zeta of FFGFT has additional zeros — generated by the $\mathbb{Z}_3$ symmetry — that lie on a different line. That is a touch, not a solution. A mountain and a river touch at the riverbed — but the mountain does not explain the river.*

The Euler Factor Threshold: Why Physics Stops

From k $\geq$ 7, primes no longer contribute physically. The Euler factors converge to 1 — they are suppressed by the fractal damping of the torus. More concretely: From p* $\approx$ 9.76 (at s=2), the product density of the Galois lattice is at random level. Beyond this point, no Galois product is distinguishable from random coincidence. [B]

This boundary does not lie in the tool — it lies in the object. This is the Weyl obstruction: There is a fundamental limit to which algebraic structures can make physical distinctions. Beyond this limit, all patterns become statistically indistinguishable.

For All Readers: *In music there is the concept of the "boundary tone" — the highest overtone still perceived as harmonious. Beyond it, everything sounds equally rough. In FFGFT there is an algebraic counterpart: The primes up to k=6 (prime 7) are physically distinguishable, from k=7 no longer. This is not a limitation of the theory — it is a property of the Galois structure.*

Connection to the Riemann Hypothesis — Without Overstepping

The full zero structure of the orbifold zeta L(s, χ_0) consists of three types: - Trivial zeros of $\zeta(s)$: at s = -2, -4, -6, ... - Non-trivial Riemann zeros: at Re(s) = 1/2 (assumed) - Orbifold zeros: at $s = 2\pi ik/\ln 3$, on $\mathrm{Re}(s) = 0$

The second category is taken over unchanged by FFGFT — neither proven nor changed. The Riemann hypothesis remains open.

What FFGFT achieves: It gives the zeta function a **physical interpretation**. The zeros of the orbifold zeta are not abstract numbers — they are spectral signatures of $\mathbb{Z}_3$ symmetry breaking on T^4. The prime hierarchy in the Euler product is not a curiosity of number theory — it is the order of the particle structure.

Mathematics and physics speak the same language. Euler knew this in 1739 when he drew the Tonnetz. FFGFT restates this proposition — more precisely, more deeply, and with predictions that can be measured.

Chapter 17

The Particle Diagram: Representation, Not Ontology

In September 2026, Johann Pascher wrote to the IPI mailing list: "On the nature of the particle diagram — representations, not ontology." That was not a rhetorical point — it was a precise methodological statement.

And it deserves its own chapter.

What Doc. 356 Shows — and What It Does Not Show

The particle overview diagram of FFGFT (Doc. 356) shows the four orbit classes f_1, f_2, f_3, f_4 with their algebraic properties: Majorana/Dirac, charge, color, generation. It is a clean, complete picture of the algebraic structure.

It shows no particles in space.

This is not an error in the diagram. It is its correct interpretation. The Galois classes f_1, f_2, f_3, f_4 are algebraic objects on a compact torus. They have no position in three-dimensional space. They have no extension. They are topological invariants — things that cannot be calculated away, no matter which coordinate system one chooses.

For All Readers: *A map shows mountains and rivers — but the line on the map is not the mountain. It is its representation. The particle diagram of FFGFT shows algebraic objects — but the Galois class f_1 is not the electron. It represents it — with a precision that no other model achieves.*

The Three Levels of Abstraction

There are three different levels on which one can speak about particles:

Level 1 — The Experiment: A detector registers energy deposits, track curvatures in magnetic fields, decay products. This is the particle as measurement event.

Level 2 — The Standard Model: A quantum field with certain symmetry properties, charges, masses. This is the particle as physical object with properties.

Level 3 — FFGFT: A Galois class in $GF(27)^*$, a fixed point of $T^4/\mathbb{Z}_3$, an orbit under the Frobenius automorphism. This is the particle as algebraic invariant of a geometric structure.

FFGFT works on Level 3 and makes precise predictions for Level 1. Level 2 — the Standard Model — is an intermediate language in which FFGFT formulates its predictions, but not in which it thinks.

For All Readers: *An architect draws a blueprint. The construction worker reads the blueprint. The house stands. The blueprint is not the house — it is a representation from which the house follows. FFGFT is the blueprint. The experiment is the house. The Standard Model is the construction worker — a useful intermediate level.*

Why the Distinction Matters

There is a question that has been asked repeatedly in the history of physics and has received different answers: What is an electron really?

Different theories give different answers. Classical electrodynamics: a charged point particle. Quantum mechanics: a state in a Hilbert space. Quantum field theory: an excitation of a field. The Standard Model: a fermion in the fundamental representation of the gauge group SU(3)×SU(2)×U(1).

FFGFT gives no new answer to the question "what is an electron really". It says: The orbit 5, 15, 19 in $GF(27)^*$ — the algebraic class f_3 — is the most precise mathematical description we have. Whether this description is identical to what the detector registers — that is a philosophical question, not a physical one.

This is not a weakness. It is methodological honesty. Physics can describe structures. It can make predictions. It can say which structures force which other structures. But "what nature really is" — that lies beyond physics.

For All Readers: *The music of Bach's fugues is completely written in notes. The notes are not the music — the music arises only in the playing, in the hearing. The algebraic invariants of FFGFT are like the notes:*

precise, complete, reproducible. The electron in the detector is the music. The identity of both — whether the note is the sounding — that is the ontological question that no theory can answer.

What FFGFT Claims — and What It Does Not

FFGFT claims: - The number of fermion families is structurally determined: nine - The Galois classes correspond to the known particles in all measurable properties - The masses, charges, mixing angles follow from the algebraic structure - The predictions are falsifiable and precise

FFGFT does not claim: - The Galois classes are the particles (ontological identification) - The $T^4/\mathbb{Z}_3$ geometry is "more real" than three-dimensional space - That there could be no deeper description

This is the position of geometric constitutionalism from Chapter 3, now concretely at the particle diagram:

Geometry forces algebra. Algebra corresponds to observation. Ontology remains open.

This modesty is not a weakness. Theories that claim more than they can prove are weak — no matter how self-confidently they appear. FFGFT proves much. And clearly states what it does not prove.

Chapter 18

Superposition Without Time

The measurement problem of quantum mechanics is one of the oldest unsolved problems in physics. Not because it would be experimentally unclear — quantum mechanics makes predictions with terrifying accuracy. But because the formalism itself has a gap: It describes what is measured, but not how the measurement happens.

At its center stands a question: What happens in superposition?

What Standard Quantum Mechanics Says — and What It Conceals

An electron can be in a superposition of two spin states: $|\psi\rangle$ = c↑$|\uparrow\rangle$ + c↓$|\downarrow\rangle$

This state is mathematically well-defined. But it has no intrinsic time scale. The question "how long is the electron in this superposition?" cannot be asked in the QM formalism — for that one would need a time operator, which does not exist (Pauli 1933).

What does exist is the coherence time T_2 — a phenomenological quantity that describes how quickly interaction with the environment destroys the superposition. That is an external, environment-dependent quantity — no intrinsic property of the state itself.

In standard quantum mechanics, superposition is **timeless**: no built-in when, no built-in how-long. [Doc. 334]

For All Readers: *Imagine you know that a stack of coins lies on a table — but you do not know how long it has been there, and the formalism you use does not allow this question. That is superposition in standard quantum mechanics: completely described in one dimension (what), completely silent in another (when and how long).*

Three Structural Consequences of Timeless Superposition

The absence of a time operator has three direct consequences that are rarely made explicit in textbooks:

First — Superposition is timeless. There is no intrinsic clock that says how long a system is in a superposition. The coherence time is a property of the environment, not of the state.

Second — Entanglement is without simultaneity. Two entangled particles share a common state without built-in time structure. The question of the simultaneity of a measurement is not a QM question — it is a question of the external parameter t, and thus a relativistic question. This is the deepest reason why QM and Lorentz invariance are compatible: because "simultaneity" stands outside the formalism.

Third — Collapse is without duration. The measurement act $|\psi\rangle \rightarrow |\psi_k\rangle$ is instantaneous in the parameter t. How long it takes is not defined in the QM formalism. This is not an inconsistency — it is a direct consequence of the absence of the time operator.

For All Readers: *In QM, the collapse occurs — the jump from superposition to the definite state — without duration, without process, without mechanism. This is not a flaw in the theory. It is a structural silence: The formalism describes what is present after the measurement, not what happens during the measurement. Like a film without the dissolve: Frame A, then Frame B, nothing between.*

What $\tilde{T} \cdot m = 1$ Changes

In FFGFT, each mode carries its own intrinsic time scale: $\tilde{T}_i = 1/m_i$

This is not a phenomenological quantity — it is a geometric property of the mode, determined by its mass. A state is its time scale. [B Doc. 334]

This has direct consequences for the three points above:

Superposition within one mode — "intermediate torsion" in the language of Doc. 174 — has a natural time scale $\tilde{T}_i$. That is structurally consistent.

Superposition across different modes — a superposition of states with $m_1 \neq m_2$ — superposes incompatible time scales $\tilde{T}_1 \neq \tilde{T}_2$. This is structurally problematic: There is no common geometric concept of time. [B]

The measurement act (Type I) has a natural direction — lossy from S^1_m to $\mathbb{R}_t$, the winding number is discarded. The arrow of time is thus no external additional assumption, but a consequence of the geometry of measurement. [K]

For All Readers: *When two musicians play different time signatures — one 3/4, one 4/4 — there is no common beat. One can still hear the tones together, but they lose the common time structure. In FFGFT this is not a metaphorical image: Particles with different masses literally have different geometric time scales. A cross-modal superposition is a tempo conflict, not a stable state.*

The Pauli Theorem and Compact Time

The Pauli theorem applies to an open time parameter conjugate to a Hamiltonian bounded below. It excludes a self-adjoint time operator in this case.

But $\tilde{T} \cdot m = 1$ realizes a different case: compact time. Time is not open — it is periodic, with period R_m = $\hbar$/(mc^2). The conjugate spectrum is then discrete: the winding numbers, the discrete mass ladder. And the Pauli theorem does not apply to compact time dimensions — because the Pauli condition (spectrum bounded below) is formulated differently for a discrete winding number spectrum. [B Doc. 307/334]

$\tilde{T} \cdot m = 1$ does not circumvent the Pauli theorem — it satisfies it in a way that Pauli did not consider. Compact time is the consistent way to integrate time into the Hilbert space.

For All Readers: *The Pauli theorem says: if you want to insert time into quantum mechanics in a certain way, it does not work. FFGFT inserts time in a different way — as a compact quantity, as a period on a circle, not as an open line. For that, the prohibition does not apply.*

What Remains Open

The measurement problem of quantum mechanics is not solved by this. FFGFT reframes it — but “reframed” is not “solved”.

What FFGFT achieves: It gives intrinsic time a geometric meaning. It gives the arrow of time a geometric reason. It makes cross-modal superposition structurally questionable — which is more precise than “somehow it collapses”.

What remains open [S]: Why and how exactly the measurement act discards the winding number. What triggers the Type I transition physically. Whether FFGFT completely solves the measurement problem or only relocates it.

This is an open bridge. Honestly declared. And more precise than most formulations of the measurement problem that claim to solve it.

Chapter 19

Quantum Operations in FFGFT

Quantum mechanics describes operations on states: unitary time evolution, measurement, entanglement, collapse. FFGFT changes none of these tools — but it changes *what they mean*. For when every mass carries its own intrinsic time scale ($\tilde{T}_i = 1/m_i$), then the "time" in $e^{-i\hat{H}t}$ is no longer an external stage, but a geometric object.

Three Operation Types on T^4

FFGFT distinguishes three logically different operations that mediate between the fundamental torus T^4 and the observable space $\mathbb{R}^3$: [K Doc. 330]

Type I — Duality decompactification. $S^1_m \to \mathbb{R}_t$ mediated by $\tilde{T} \cdot m = 1$. A compact mass dimension is unrolled into a non-compact time dimension. The operation is *information-preserving* — a bijective embedding, no projection. What is discrete on the mass side (Galois orbits) appears on the time side as a continuous spectrum of intrinsic scales.

Type II — Geometric projection. $D3 \to D2 \to D1$: the holographic projection from four to three space dimensions. This operation is *lossy* — not invertible. It generates confinement: states that are not $\mathbb{Z}_3$-invariant under the projection (single quarks) do not arrive on the observer side. Confinement is not a force law, but a projection filter.

Type III — Representational translation. $T^4 \leftrightarrow \mathscr{H}$ (Hilbert space/-matrix): the translation between the geometric description and quantum mechanics. This operation is *bijective and lossless* — a change of language, no dimension reduction. The quantum state $|\psi\rangle$ represents geometric modes of the torus; operators on $\mathscr{H}$ are images of symmetries on T^4.

For All Readers: *Imagine a landscape that can be mapped in three different ways: as a three-dimensional model (the torus), as a two-dimensional map (the projection into observable space), and as a table of elevation values (the Hilbert space). The landscape is the same. But the map loses elevation information — that is Type II, and that is why quarks are confined. The table loses nothing — that is Type III, and that is why quantum mechanics and FFGFT make the same predictions. The unrolling of the map into a timeline is Type I — and that is the meaning of $\tilde{T} \cdot m = 1$.*

Superposition: Not Timeless, But Scale-Dependent

Standard QM treats superposition as a timeless mathematical object: $c_1 |\psi_1\rangle + c_2 |\psi_2\rangle$ exists at a fixed t, without intrinsic duration. The Pauli theorem shows why: for a Hamiltonian bounded below, there is no self-adjoint time operator. Time stands outside the formalism. [B Doc. 307]

In FFGFT, each mode receives an intrinsic time scale: $\tilde{T}_i = 1/m_i$. Superposition of two modes with different masses $m_1 \neq m_2$ is therefore not timeless — it has a geometrically defined coherence scale on which the interference is stable or unstable. The decoherence time is not imposed from outside, but follows from the geometry:

$$\tau_{\text{koh}} \sim \frac{1}{|m_1 - m_2|}$$

Two states with very different masses decohere quickly — algebraically forced, not phenomenologically set. [S Doc. 334]

Entanglement and Non-Locality

Entanglement in FFGFT is a correlation between geometric modes on T^4. Since T^4 is compact, there is no fundamental locality in the sense of Minkowski space — the concept of "spatial separation" arises only through the Type II projection into $\mathbb{R}^3$.

This means: Entanglement violates the Bell inequalities in FFGFT for the same reason as in standard QM — the state space is not locally realistic. But FFGFT gives this a geometric meaning: the non-locality of entanglement reflects the global topology of the torus. Entangled modes are global modes of T^4, not separate objects at different locations. [S Doc. 334]

Collapse and Measurement

The measurement problem of quantum mechanics — why does $|\psi\rangle$ collapse to an eigenstate upon measurement? — is in FFGFT a question of Type II projection. A measurement is an interaction that forces the measured state into observable space. Only $\mathbb{Z}_3$-invariant modes survive the projection — these are the observable states. What appears as "collapse" is the selection by the projection filter: non-invariant superposition is not observed, not annihilated. [S Doc. 330]

For All Readers: *Imagine a river flowing through a narrow sieve. The water as a whole continues to flow — but the sieve lets only certain particles through. What arrives on the other side looks like a "collapse" to discrete objects. The sieve is the geometric projection of T^4 onto $\mathbb{R}^3$. The quantum states continue to flow — but the observer sees only what fits through the sieve.*

Deterministic Logic and the Extended Schrödinger Equation

Two points go beyond a mere reinterpretation of the standard tools. FFGFT changes not only what the quantum operations *mean* — at two places it gives them a different form.

First: the Schrödinger equation is extended. In standard QM we have $i\hbar\,\partial_t\psi = \hat{H}\psi$ with a global time t, the same everywhere. In FFGFT time is a field $T(x,t) = 1/m(x,t)$ — at every point as long as the local mass dictates. The equation thereby becomes: [S Doc. 020]

$$i\hbar\, T(x,t)\,\frac{\partial\psi}{\partial t} = \hat{H}_0\,\psi + \hat{V}_{T0}(x,t)\,\psi, \qquad \hat{V}_{T0} = \xi\hbar^2\,\frac{\delta E(x,t)}{E_{\mathrm{Pl}}}$$

The factor $T(x,t)$ in front of the time derivative is the local time scale; the additional term $\hat{V}_{T0}$ is the coupling to the time field itself, of order $\xi \approx 1.33 \cdot 10^{-4}$. In the limit $T \to$ const, $\xi \to 0$ the equation falls back exactly onto the standard form. What FFGFT adds is therefore no break with the Schrödinger equation, but a small, geometrically determined correction term — and a time field in place of the time constant.

Second: the logic is deterministic. Standard QM describes a qubit as a vector $\alpha|0\rangle + \beta|1\rangle$ in which $|\alpha|^2$ and $|\beta|^2$ are *probabilities* — chance is built in. FFGFT describes the same qubit as a triple (z,r,θ) with $z^2 + r^2 = 1$: z is the projection onto the computational basis, r the superposition amplitude, θ the phase. The decisive difference: r^2 is not a probability but the *energy density* of the superposition state. [S Doc. 147]

The logic functions thus become deterministic maps on this triple. The Hadamard gate, for instance, reads:

$$H_{T0} \;:\; (z,r,\theta) \longmapsto \left(re^{-\xi \ln n/D_f},\; ze^{-\xi \ln n/D_f},\; \theta + \tfrac{\pi}{2}\right)$$

The swap $(z,r) \to (r,z)$ is the basis change; the exponential factor is a ξ-small damping that stabilises multi-qubit entanglement; $D_f = 3-\xi$ is the fractal dimension. No dice, no amplitude that "randomly" turns into a result — only a map that can be run backwards.

And measurement? It drives the system deterministically into the nearest stable pole of the state space — depending on the exact phase state before. The "probability" of standard QM describes, in this picture, our *ignorance* of that phase state, not a fundamental randomness of nature. [S Doc. 175]

For All Readers: *A die on the table is not random. If you know the throw height, spin, air resistance and table surface exactly, you know the result in advance. We do not know them — that is why we say "probability one sixth". FFGFT says: the qubit is that die. Quantum probability is real as a description of our knowledge — but not as a property of the die.*

Third: the Bell inequality is extended. Bell's inequality places a bound on every locally realistic theory: $|E(a,b) - E(a,c)| + |E(a',b) + E(a',c)| \leq 2$. Quantum mechanics violates it up to the Tsirelson bound $2\sqrt{2}$ — and that is commonly taken as proof that nature is not locally realistic. FFGFT starts differently here. The correlations do not arise through action at a distance, but through a *local correlation field* propagating causally at c — the two particles share a common torsion node on T^4. That changes the inequality in two ways: [S Doc. 023/147]

$$|E(a,b) - E(a,c)| + |E(a',b) + E(a',c)| \ \le \ 2 + \epsilon_{T0}, \qquad \epsilon_{T0} = \xi \frac{2\langle E\rangle \ell_P}{r_{12}}$$

The right-hand side is no longer 2 but $2 + \epsilon_{T0}$, with a time-field term depending on the particle separation r_{12}. And the correlation function itself carries a ξ damping:

$$E_{T0}(a,b) = -\cos(a-b)\left(1 - \xi f(n,l,j)\right), \qquad S_{T0}(n) = 2\sqrt{2}\, e^{-\xi \ln n/D_f}$$

For n entangled qubits the CHSH value thus lies $\xi \ln n/D_f$ *below* the Tsirelson bound — not above it. This is the point where FFGFT and standard QM separate measurably: QM predicts exactly $2\sqrt{2}$, FFGFT exactly $2\sqrt{2}$ minus a tiny, geometrically fixed amount.

What makes this testable. The deterministic representation reproduces all interference effects of standard QM — otherwise it would be wrong. The difference lies in the ξ deviation of the Bell correlation. Measurements on real quantum hardware (May 2026) confirm the Bell violation and that the FFGFT circuits run ($S = 2.74$, i.e. $96.9\,\%$ of the bound) — but the ξ effect itself still lies far below today's noise and cannot be resolved with current hardware. [K Doc. 147] The prediction stands; the experiment that decides it does not yet exist.

What FFGFT Changes and What It Does Not

FFGFT does not break with quantum mechanics. In the limit $\xi \to 0$, $T \to$ const all computational rules — Born rule, Schrödinger equation, unitarity — fall back exactly onto the standard form. What FFGFT changes is twofold: the ontological status of these rules, and at two places their form:

- The Schrödinger equation carries a time field $T(x,t)$ and a ξ-small coupling term — an extension, not a replacement.
- The logic functions are deterministic maps on (z,r,θ); r^2 is energy density, not probability. Quantum randomness is epistemic, not ontic.
- The Bell inequality carries a time-field term $2 + \epsilon_{T0}$ and the correlation a ξ damping; non-locality is replaced by a local correlation field propagating at c.

- Unitary time evolution is Type III translation: a change of representation, no physical process on T^4.
- Masses are no free parameters, but Galois orbits. Thus the time scales of time evolution are also geometrically determined.
- Collapse is Type II projection: selection by invariance, no mysterious jump.
- Entanglement is global topology, no spooky action at a distance.

What quantum mechanics describes phenomenologically has a geometric cause in FFGFT. That is less than a new quantum theory — and more than an interpretation.

Chapter 20

Geometry as the Deepest Foundation

Every theory has a foundation. Something that is not further explained — that is set. An axiom, a setting, a beginning.

In Newtonian mechanics that is absolute space and absolute time. In relativity theory the metric of spacetime. In the Standard Model the gauge group SU(3)×SU(2)×U(1) and the list of fields.

In FFGFT it is geometry: $T^4/\mathbb{Z}_3$.

What "Deepest Foundation" Means

The question "what is the deepest foundation of a theory" is not only philosophical — it is physically relevant. For from the foundation depends what is explained and what is not.

In the Standard Model, the gauge group is set. From it follow the conservation laws, the couplings, the symmetries. What does not follow: Why this group. Why these particles. Why these masses.

In FFGFT, the geometry $T^4/\mathbb{Z}_3$ is set [SETZUNG]. From it follow algebraically the Galois fields, the fixed points, the orbit structures. From these follow the particles, their masses, their couplings — provably, without further degrees of freedom.

What does not follow: Why $T^4/\mathbb{Z}_3$. That is the fundamental question that every physical theory ultimately leaves open.

For All Readers: *Every building stands on a foundation. The foundation itself stands on the ground. The ground stands on the earth's crust, which stands on the mantle — and eventually one asks: why does the earth stand at all? FFGFT answers many "why" questions — why nine particles, why these masses, why these constants. But the last question — why*

$T^4/\mathbb{Z}_3$ — remains open. That is not a weakness. Every honest theory has such a last question.

The Constitution Type: Geometry → Algebra → Empirics

In an IPI conversation in September 2026, FFGFT was described as "empirically constituted" — because it reproduces measured values. That is a mislabeling. The correct description:

Geometric [SETZUNG] → Algebraic consequences [B] → Independent empirical comparison [K]

This is the decisive difference. FFGFT does not take measured values as input and then seeks a theory that reproduces them — that would be empirical constitution. FFGFT sets a geometry, derives algebraically necessary structures from it, and then independently compares with measurements.

The measured values come at the end — as a test, not as input.

For All Readers: *An architect designs a building according to aesthetic and static principles. Then he checks whether it meets the building regulations. He does not design backwards — he does not read the building regulations and then ask which building they just barely allow. Geometric constitution is the design. Empirical comparison is the building inspection.*

Why Geometry Is the Deepest Foundation

There are theories that have algebra as foundation — finite groups, Galois fields, algebraic structures. And there are theories that have geometry as foundation — Riemannian manifolds, fiber bundles, orbifolds.

FFGFT sets geometry as the deepest foundation — and derives algebra from it. This is a statement about the order of explanation.

Why is geometry deeper than algebra? Because geometry forces local consistency. I can postulate an algebraic structure — I can simply say "let there be $GF(27)^*$". A geometric structure must be consistent — the torus must be locally flat everywhere, the $\mathbb{Z}_3$ operation must be defined everywhere, the fixed points must remain invariant under the operation. Geometry has continuity requirements that algebra does not have.

This is the reason why discrete lattice models like HLV fail: They set algebraic structures (lattice points, distances) without the geometric consistency condition. The consequence are X4 overlaps — places where the structure contradicts itself. [B Doc. 330]

For All Readers: *You can write on paper: "I have a triangle with three angles of 90° each." Algebraically that is definable. But geometrically it* *is impossible — three right angles in a flat triangle contradict the angle sum of 180°. Geometry forces consistency. Algebra does not. That is why geometry is the deeper foundation.*

Lorentz Invariance: Locally Unrestricted, Globally Topological

A frequent question about FFGFT: Does the compact geometry of the torus not violate Lorentz invariance? Is a compact space not by definition distinguished — does it not have preferred directions?

The answer is precise: No — locally. Yes — globally. [K Doc. 312]

Locally — at every point of the torus — the metric is the same as in flat Minkowski space. No local observer can tell that they live on a torus. Local Lorentz invariance is fully preserved.

Globally — over the entire torus — there are topological distinctions: winding numbers, fixed points, homotopy classes. These global structures are the foundation of the particle structure. They violate no local symmetry — they add global topology.

This is the difference between a sheet of paper (locally flat, globally open) and a torus (locally flat, globally closed). An ant on the torus cannot locally distinguish whether it lives on a sheet or a torus. But the global structure decides which paths are possible.

For All Readers: *When you stand in a room, you see four walls, a ceiling, a floor. Locally it looks like a normal room. Now imagine the walls are "periodic" — when you walk through the right wall, you come back in on the left. That is a torus room. Locally: normal. Globally: different. FFGFT lives in such a room. And the global structure — the periodicity, the windings, the fixed points — that is the particle structure.*

Bell Correlations: Real, But Geometrically Structured

A final consequence of the geometric foundation: Bell correlations — the non-classical correlations of entangled particles — are treated in FFGFT as real but geometrically structured. [B Doc. 230]

They are not ontologically nonlocal in the sense of "signal propagates faster than light". They arise from holonomy on the torus — the phenomenon that a vector transported around a closed curve on a curved or compact space returns rotated.

Bell correlations as holonomy — that is a statement about the geometric deep structure, not about a mysterious action at a distance. It is [B] for the group theory behind it, and [S] for the complete physical elaboration.

The Position Summarized

The deepest foundation of FFGFT is geometry — $T^4/\mathbb{Z}_3$. From it follows algebra. From it follow predictions. From it follows comparison with empirics.

This is not a cycle. It is a chain — with a beginning (geometry), a middle (algebra), and an end (experiment). The beginning is set [SETZUNG]. The middle is proven [B]. The end is tested [K].

And the question of why the beginning is as it is — that remains open. As with every honest theory.

Chapter 21

What Is Proven, What Is Sketched, What Remains Open

Physics is not an enumeration of truths. It is a process — an ongoing dialectic between theory and experiment, between proof and conjecture, between what is known and what is not yet known.

FFGFT documents this process with a system that is unusual in theoretical physics: Every statement carries its epistemic status. Nothing is asserted without marking what it is.

This chapter lays out the complete balance.

The Status System — Once More in Full

Four markers structure the entire corpus:

[SETZUNG] — The axiom. The geometric foundation $T^4/\mathbb{Z}_3$ is not derived — it is set. That is not a weakness: Every theory needs a beginning. The beginning is transparently declared here, not hidden.

[B] — Algebraically proven. What follows from the axiom with mathematical necessity. Independent of free parameters, independent of measurements. Proofs of this class are verified with test scripts — reproducible, auditable.

[K] — Derived from ξ, numerically tested. Quantitative predictions that have been compared with measured values. The comparison is transparent: anchor, steps, corrections and extraction method are documented.

[S] — Sketched. Structurally plausible, consistent with the corpus, but the complete derivation is missing. That is not an excuse — it is an invitation: Here lies the next task.

[X] — Excluded. Statements that have been refuted in the corpus — by algebraic proof or by test script refutation.

[Q] — Source. External primary literature, correctly cited. Not FFGFT's own derivation.

What Is Proven [B]

From $T^4/\mathbb{Z}_3$ follow algebraically, without further assumptions:

Nine fixed points — the foundation of the fermion structure. Three generations as eigenvalues of the $\mathbb{Z}_3$-circulant. Prohibition of the fourth generation. Eight gluons from $GF(27)^*$ three-orbits. Frobenius separation massive/massless. Majorana character of ν_1, ν_2. Dirac character of ν_3. Charge quantization Q $\in$ 0, ±1/3, ±2/3, ±1 from the Legendre symbol modulo 13. Gell-Mann-Nishijima relation Q = I_3 + Y/2 from the jE7 projector structure. Prohibition of electrically neutral colored fermions. Coupling matrix $f_3 \times f_4$ as permutation matrix. Entries $1/\sqrt{2}$ in $f_1 \times f_3$. Prime forcing 2,3,5,7,11,13 by 3^k–1. Hierarchical Euler product of the orbifold zeta. Orbifold zeros on Re(s)=0. All Yukawa coefficients in the Galois lattice. Self-adjointness of $\hat{F}$. Type III pullback as exact isometry. Cross-modal superposition generates incompatible time scales.

What Is Derived and Measured [K]

Quantitatively, with comparison values:

Prediction	FFGFT	Experiment	Dev.
m_μ/m_e (bare)	207.846	206.768	0.52%
$(m_\mu/m_e)^2$ = 43200	exact	42753	1.0%
$1/\alpha$ = 3700/27	137.037	137.036	7.6 ppm
$\sin^2\theta_W$	0.2305	0.23122	0.06%
Δm^2_{atm}	2.476 × 10^{-3} eV²	2.500 × 10^{-3} eV²	1.0%
Δm^2_{sol}	7.565 × 10^{-5} eV²	7.530 × 10^{-5} eV²	0.46%
Σm_ν	64.3 meV	< 120 meV	consistent
m_τ prediction	1776.78 MeV	1776.86 ± 0.12 MeV	0.4σ
Koide scalar Q	0.6677	2/3 = 0.6667	0.04%

What Is Open [S]

The open bridges of the corpus — precisely formulated, not concealed:

R113 bridge: Why do the lepton residuals ε_i fall with generation? The residuals are calculated [K] — their attribution to QED extraction and SI projection is conjecture [S].

Koide amplitude: d/c = $\sqrt{2}$ derived from $\mathbb{Z}_3$ uniform distribution [B Doc. 353].

τ_μ directly from FFGFT — the only convention-free test of the weak sector, without detour via G_F and v.

Cosmic exponent: Why level 10 of the ξ ladder for the cosmological constant? It decomposes the 10^{123} fine-tuning into two algebraic contributions — but the exponent 10 itself is not yet derived forward (P20).

CMB peak selection: The sequence 1, 6, 14, 26 of cosmic microwave background maxima generated from the torus structure.

Quark/hadron sector: Complete derivation of the quark coefficients beyond the Galois lattice. The Yukawa coefficients lie in the lattice [B] — their numerical values are partly lattice-QCD calibrated, not purely geometric.

Gravity and quantum gravity: G follows from ξ [K]. But the question of whether FFGFT implies or excludes a quantum gravity is structurally clarified [B] — a separate quantum gravity does not arise because $T^4/\mathbb{Z}_3$ is pre-ordered.

What Is Excluded [X]

Not everything in the corpus is progress — some is retraction. That is just as important.

Neutrino part of Doc. 006: The "direct method" E_i = 1/ξ_i is no calculation — it produces inverted hierarchy and MeV values that were measured values, not predictions. [X Doc. 109]

QED self-energy as cause of the lepton residuals: Excluded at 112σ. [K → X as explanation]

Universal bit values (Matzke): Bit values are system-dependent — E_bit = $\hbar$c/L, Landauer's kTln2 only special case. [B Doc. 329]

The Methodological Statement

The corpus as of September 2026 has over 360 documents, 115 register entries, dozens of test scripts. It has grown — not linearly, but iteratively, with corrections, retractions, refinements.

That is not a weakness. That is science.

A theory that is never corrected is never tested. A theory that names open bridges knows where it stands. A theory that excludes what it refutes is more honest than one that conceals it.

FFGFT is a living theory — in the best sense.

Chapter 22

The Matrix Question Answered

We started at the beginning of this book with a question: Do we live in a matrix?

It is time to answer it.

What the Question Really Asks

Bostrom asks: Is someone computing us? Tegmark asks: Are we a mathematical structure among many? Zuse and Wolfram ask: Does nature run on discrete rules?

All three questions have something in common: They ask about the *relationship* between mathematics and physics. And they give different answers:

Bostrom: Mathematics is the tool of a simulator. Tegmark: Mathematics is the universe itself, one among many. Zuse/Wolfram: Mathematics is the algorithm that computes the universe.

FFGFT gives a fourth answer. One that is more precise than all three.

The FFGFT Answer

We do not live in a simulation — because a simulation presupposes a simulator, an external instance, a hardware. FFGFT knows no external instance.

We do not live in a randomly selected mathematical structure among infinitely many — because that does not explain why this structure. It is a philosophical position, not a physical theory.

We do not live in a cellular automaton or a discrete lattice — because these fail at local consistency. Discrete structures without geometric foundation generate X4 overlaps, self-contradictions. Geometry forces consistency. Algebra alone does not.

What instead:

The universe has an algebraic deep structure — and this structure allows no other particles than those we know.

This is more than Bostrom. Bostrom only asks whether there is a simulator. FFGFT asks what the structure of reality forces — and gives an answer that is falsifiable, algebraically proven, and numerically tested.

This is more than Tegmark. Tegmark says all mathematical structures exist. FFGFT says the structure $T^4/\mathbb{Z}_3$ forces exactly these particles, exactly these constants — and excludes all others. This is not a selection from a set — it is an algebraic prohibition.

This is more than Zuse. Zuse says nature computes on discrete rules. FFGFT says nature has a geometric foundation from which algebraically follows what must follow. No computation, no external instance, no rules from outside — only inner algebraic necessity.

For All Readers: *Three pictures for the three alternatives: Simulation — the universe is like a computer game. Someone plays it. Tegmark — the universe is like a number. It exists because it is mathematically possible. Zuse — the universe is like an algorithm. It runs itself. FFGFT — the universe is like a proof. It cannot be otherwise.*

What the Proof Shows

This is the statement that this book has prepared from beginning to end:

The algebra of $GF(27)^*$ forces exactly four orbit classes. Each class has three elements — three generations. Four classes, three generations: nine fermion families. Not eight. Not ten.

A tenth family has algebraically no room. Not because no one finds it. Because the structure forbids it.

This is not a model. This is a proof.

What Remains Open — and Why That Is Good

The open bridges of this book — R113, P20, the quark sector, the Koide amplitude — are not signs of failure. They are signs of precision.

A theory that explains everything explains nothing. It is too soft, too flexible, too ready to encompass every observation post hoc. That is not physics — that is rhetoric.

FFGFT explains much — and precisely names what it does not explain. That is the strongest form of honesty a theory can have.

And every open bridge is a concrete next step. The physics of the coming decades will decide:

Is ν_3 a Dirac neutrino? The neutrinoless double beta decay experiment will answer.

Is $\Sigma m_\nu \approx$ 64 meV? KATRIN and Euclid will measure.

Does m_τ lie at 1776.78 MeV? Belle II and BESIII will become more precise.

Is there a fourth quark generation? The LHC continues to search — FFGFT says what it will not find.

Each of these questions has a clear answer that follows from the theory. And each wrong answer from experiment would show where the theory must be corrected. That is science — not simulation, not ensemble, not algorithm.

The Last Sentence

Feynman wrote 1/137 on the wall and asked: Why?

FFGFT writes: Because 3700/27. And 3700 and 27 are group orders of the Galois field hierarchy of $T^4/\mathbb{Z}_3$.

That is not a complete answer to Feynman's question. But it is a deeper answer than any given before FFGFT.

And "deeper" is, in physics, always a step forward.

The universe is not a simulation. It is an algebraic structure — and this structure allows no other particles than those we know. That is less than ontology. And more than physics had before.

Appendix A — The Status System and Document Numbering

This book uses status markers and document references in the main text, which appear in the margin or in parentheses. This appendix explains both completely.

Why a Status System?

In theoretical physics, it is customary not to distinguish asserted results from proven results. Both appear in the same running text, often with the same tone. This makes theories difficult to audit — one does not know what is set, what is proven, and what is merely conjectured.

FFGFT uses an explicit status system that classifies every statement. This enables three things: reproducibility (every proof can be rechecked), honesty (conjectures are not presented as proofs), and progress documentation (when a sketch becomes a proof, the marker changes).

The Six Status Markers

[SETZUNG] — Axiom. Not derived, but set. In FFGFT there is exactly one setting: the geometry $T^4/\mathbb{Z}_3$ as fundamental space. Without this setting there is no theory — with it everything else follows. This is transparently declared, not hidden.

Example: "$T^4/\mathbb{Z}_3$ as fundamental geometric structure [SETZUNG]"

[B] — Proven / Algebraically proven. Follows necessarily from the setting, with mathematical proof verified with test scripts. No free parameter, no fit, no discretion. Anyone who doubts can run the test script — the result is reproducible.

Example: "$GF(27)^*$ has exactly four primitive orbit classes with three elements each [B Doc. 348]"

[K] — Confirmed. Derived from ξ and compared with measured values. The derivation is complete, the comparison is transparent: which anchor was used, which corrections were applied, which extraction method the PDG value had.

Example: "$\Delta m^2_{\text{atm}} = 120 \cdot m_\nu^2$, deviation 1.0% [K Doc. 340]"

[S] — Sketched / Speculative. Structurally plausible, consistent with the corpus, but the complete derivation is missing. That is not an excuse — it is a precise statement about the state of the work. Open bridges are [S] and registered in Doc. 190.

Example: "The falling structure of the lepton residuals ε_i could follow from the Galois recursion [S Doc. 113]"

[X] — Excluded. Statements that have been algebraically refuted in the corpus or falsified by test script. This is just as important as positive results — a theory that documents its own errors is stronger than one that conceals them.

Example: "QED self-energy as cause of the lepton residuals [X Doc. 352]"

[Q] — Source. Statements from external primary literature, correctly cited. PDG measured values are [Q]. Standard formulas of quantum field theory are [Q]. This distinguishes FFGFT's own derivations from adopted knowledge.

Example: "m_e = 0.51099895 MeV [Q, PDG 2024]"

[E] — Established. Known physical facts that are confirmed in the corpus but not newly derived. Used sparingly.

How the Markers Appear in the Book

In the main text, the markers appear as marginal notes — in the margin, not in the running text. Expert readers immediately see status and document reference. Lay readers read through without being disturbed. The appendix is the reference for those who want to know more.

Document Numbering

The FFGFT corpus consists of over 360 numbered documents (Doc. 001–360+) and an A series (A001–A284).

Doc. 001–360+ — The chronological series. Each document treats a specific topic: Doc. 006 particle masses from winding numbers, Doc. 190 the ongoing correction register, Doc. 282 the $\mathbb{Z}_3$-circulant and mass operator, Doc. 338 the Galois mass comparison, Doc. 340 the neutrino mass hierarchy, and so on. The numbers are historical — they reflect the order of creation, not the logical structure.

The A Series — The canonical, subject-organized version. 48 documents (A001–A284), thematically ordered instead of chronologically. For readers who want to systematically explore the corpus, the A series is the entry point.

Doc. 190 — The correction register. An append-only document that documents all corrections, refinements, and open bridges since the project's beginning. Entries K (corrections), P (refinements), and R (revisions) with continuous numbering. As of September 2026: R1–R115, K1–K63, P1–P44. Every open bridge is registered here — with status, reference, and the document that is supposed to close it.

Test Scripts — For every algebraic proof there is a Python test script in the directory `2/python/`. These scripts are not derivation tools — they verify the claims of the associated document against explicitly formulated assertions. Anyone who doubts can run them.

Access — All documents are freely accessible:
GitHub: https://github.com/jpascher/T0-Time-Mass-Duality/
Zenodo: DOI 10.5281/zenodo.17390357 (current release)
ORCID of the author: 0009-0000-6518-4064

Appendix B — Mathematical Foundations

This appendix summarizes the mathematical structures used in the main text. It is not a complete introduction — it gives definitions and core statements for readers who want to go deeper.

Galois Fields (Finite Fields)

A Galois field $GF(q)$ is a finite field with exactly q elements. It exists exactly when $q = p^n$ for a prime p and $n \geq 1$.

$GF(3)$: $\{0, 1, 2\}$ with addition and multiplication modulo 3. $GF(9)$: Extension of $GF(3)$ by an element i with $i^2 + 1 = 0$ (in $GF(3)$: $i^2 = 2$). $GF(27)$: Extension of $GF(3)$ by an element α that satisfies an irreducible cubic polynomial.

The multiplicative group $GF(q)^* = GF(q) \setminus \{0\}$ is cyclic of order $q - 1$. For $GF(27)^*$: order $26 = 2 \times 13$.

The Frobenius Automorphism

On GF(p^n), the Frobenius automorphism is defined by: $\varphi : x \mapsto x^p$

It is a field automorphism — it preserves addition and multiplication. Its order is n: φ^n = id. On $GF(27)$: $\varphi : x \mapsto x^3$, order 3.

The fixed points of φ on $GF(27)^*$ are the elements with $x^3 = x$, i.e. $x(x^2 - 1) = 0$: the elements $\{+1, -1\}$ (plus 0). These correspond to the massive sector.

Orbits and Three-Orbits

An orbit of x under φ is the set $\{x, \varphi(x), \varphi^2(x), ...\}$ until repetition. On $\mathbb{Z}_{13}$ under $\varphi : x \mapsto 3x$ mod 13, four three-orbits arise: $\{1, 3, 9\}$, $\{2, 5, 6\}$, $\{4, 10, 12\}$, $\{7, 8, 11\}$.

The Legendre Symbol

The Legendre symbol (a/p) for a prime p and integer a with p $\nmid$ a is defined as:

$$(a/p) = \begin{cases} +1 & a \text{ quadratic residue mod } p \\ -1 & a \text{ quadratic non-residue mod } p \end{cases}$$

For p = 13: The quadratic residues are 1, 3, 4, 9, 10, 12.

The $\mathbb{Z}_3$-Circulant

A $\mathbb{Z}_3$-circulant is a 3×3 matrix of the form:

C = [[a, b, c], [c, a, b], [b, c, a]]

It is invariant under cyclic permutation of rows. Hermitian $\mathbb{Z}_3$-circulants (a real) always have the same three eigenvectors: ω^k = (1, ω^k, ω^{2k})/$\sqrt{3}$ for k = 0, 1, 2, where $\omega = e^{2\pi i/3}$.

The three eigenvalues are: λ_k = a + b·ω^k + c·ω^{2k}

These three eigenvalues correspond to the three lepton masses. Their calculation from the winding numbers is the core content of Doc. 282.

The $T^4/\mathbb{Z}_3$ Orbifold

T^4 is the four-dimensional torus: $T^4 = \mathbb{R}^4/\mathbb{Z}^4$ (identification of points differing by integer vectors).

$\mathbb{Z}_3$ acts on T^4 by simultaneous rotation by 120° in two complex planes. The fixed points of this action — points mapped onto themselves — form a finite set.

$T^4/\mathbb{Z}_3$ is the orbifold: the quotient space in which $\mathbb{Z}_3$-equivalent points are identified. At the fixed points, $T^4/\mathbb{Z}_3$ has a conical singularity.

The number of fixed points: 9. This follows from the Lefschetz fixed point theorem applied to the $\mathbb{Z}_3$ action on T^4.

The Zeta Function of the Torus

The spectral zeta function of the Laplace operator on $T^4/\mathbb{Z}_3$ is:

$$\zeta_{T^4/\mathbb{Z}_3}(s) = L(s, \chi_0) = (1 - 3^{-s})\zeta(s)$$

where $\zeta(s)$ is the Riemann zeta function and χ_0 is the principal character modulo 3. This L-function has: - The trivial zeros of $\zeta(s)$ at s = −2, −4, −6, ... - The non-trivial Riemann zeros at Re(s) = 1/2 (assumed) - Orbifold zeros at $s = 2\pi ik/\ln 3$ on $\mathrm{Re}(s) = 0$

$\tilde{T}\cdot$ m = 1 as Compactification Relation

On T^4, one of the four circles carries both time and mass structure. The compactification of the circle with radius $R_m = \hbar/(mc^2)$ gives:

$\tilde{T} \cdot m = \hbar/c^2$ (in natural units: $\tilde{T} \cdot m = 1$)

This is no additional assumption — it is the compactification condition of the mass circle. In natural units ($\hbar$ = c = 1), intrinsic time is the inverse of mass.

ξ = 4/30000 — The Galois Derivation

ξ follows from the identity:

$(r_\mu/r_e)^2/\xi = |GF(9)^*|^2 \cdot 5^2 \cdot |GF(27)| = 64 \cdot 25 \cdot 27 = 43200$

With r_μ/r_e = (16/5)/(4/3) = 12/5:

$\xi = (12/5)^2/43200 = (144/25)/43200 = 1/7500 = 4/30000$

The winding numbers (r_μ, r_e) come from torus topology, the Galois group orders from the algebraic structure. Both sides of the equation are algebraically forced — ξ is their quotient.

Appendix C — Key Documents (Selection)

The following list names the most important documents referenced in the main text. The complete directory is available in the GitHub repository under README.md and README_en.md.

Doc.	Title	Content
006	Particle masses	Yukawa ladder, winding numbers (r_i, p_i)
011	Fine-structure	α from ξ and E_0
077	E=mc^2 = E=m	c as convention, not constant of nature
180	L_0 derivation	Derivation chain $\xi \to G \to L_0$
190	Correction register	All K/P/R entries, open bridges
241	Two layers	Ratio level vs. SI anchor point
258/259	Koide scalar	Q_FFGFT = 0.6677 from Galois structure
282	$\mathbb{Z}_3$-circulant	Mass operator on $\mathbb{C}^3$ fiber
307/334	Time in QM	Pauli theorem, compact time, superposition
312	Final scale	$\Lambda \to 1/R_H^2$, 10^{123} decomposition
317	Lepton masses	Winding numbers from Galois fields
325	Hawking	KMS from $\tilde{T} \cdot m = 1$, $\mathbb{Z}_3$ triality selection

Doc.	Title	Content
327	Self-adjointness	$\hat{F}$ self-adjoint, ξ = λ_min [B]
330	T^4 operations	Three operation types Type I/II/III
333	K_frak/Duality	Two meanings of "unrolling"
336	$GF(9)$↔FFGFT	Frobenius = FFGFT sector pairing [B]
338	Galois masses	$(m_\mu/m_e)^2$ = 43200 both sides [K]
339	Frobenius separation	Massive/massless in $GF(27)^*$ [B]
340	Neutrino Galois	Δm^2_{atm}, Δm^2_{sol} from $GF(27)^*$ [K]
341	$GF(27)$ in GALG	FFGFT↔GALG algebraic comparison
342	Factorization	Prime harmonics, Galois classes [B]
343	Zeta function	$\zeta_{T^4/\mathbb{Z}_3}(s) = (1-3^{-s})\zeta(s)$ [B]
346	Charge quantization	QR/NQR mod 13 → Q ∈ 0,±1/3,±2/3,±1 [B]
347	Gell-Mann-Nishijima	$Q = I_3 + Y/2$ algebraically [B]
348	Generation orbits	Coupling matrix, 1/$\sqrt{2}$ entries [B]
351	PDG model-dependent	Measurement extraction
352	Lepton residuals	ε_i in units ξ, QED excluded [K]
358	Harmonics/Algebra	All Yukawa coefficients in Galois lattice [B]

Appendix D — Open Bridges (as of September 2026)

The following open bridges are registered in the correction register Doc. 190. They define the next steps of work.

R113 Bridge [S] — Why do the lepton residuals ε_i fall with generation? The residuals are calculated [K], their cause is attribution [S]. Only surviving candidate: Proportionality to Δn_θ (Galois recursion). Falsifiable prediction: m_τ = 1776.78 MeV.

τ_μ **directly** [S] — Muon lifetime from FFGFT without detour via G_F and v. The only convention-free test of the weak sector (R112).

Cosmic exponent P20 [S] — Why level 10 of the ξ ladder for $\Lambda *$? The 10^{123} problem decomposes into two algebraic contributions (77.5 + 44.8 decades) — but the exponent 10 is not derived forward.

CMB peaks [S] — Sequence 1, 6, 14, 26 of cosmic microwave background maxima generated from the torus mode structure.

Quark/hadron sector [S] — Complete derivation of the quark coefficients. The Yukawa coefficients lie in the Galois lattice [B], their exact values are partly lattice-QCD calibrated.

2-loop corrections to α_s(M_Z) [S] — Strong coupling constant at two-loop level from the Galois structure.

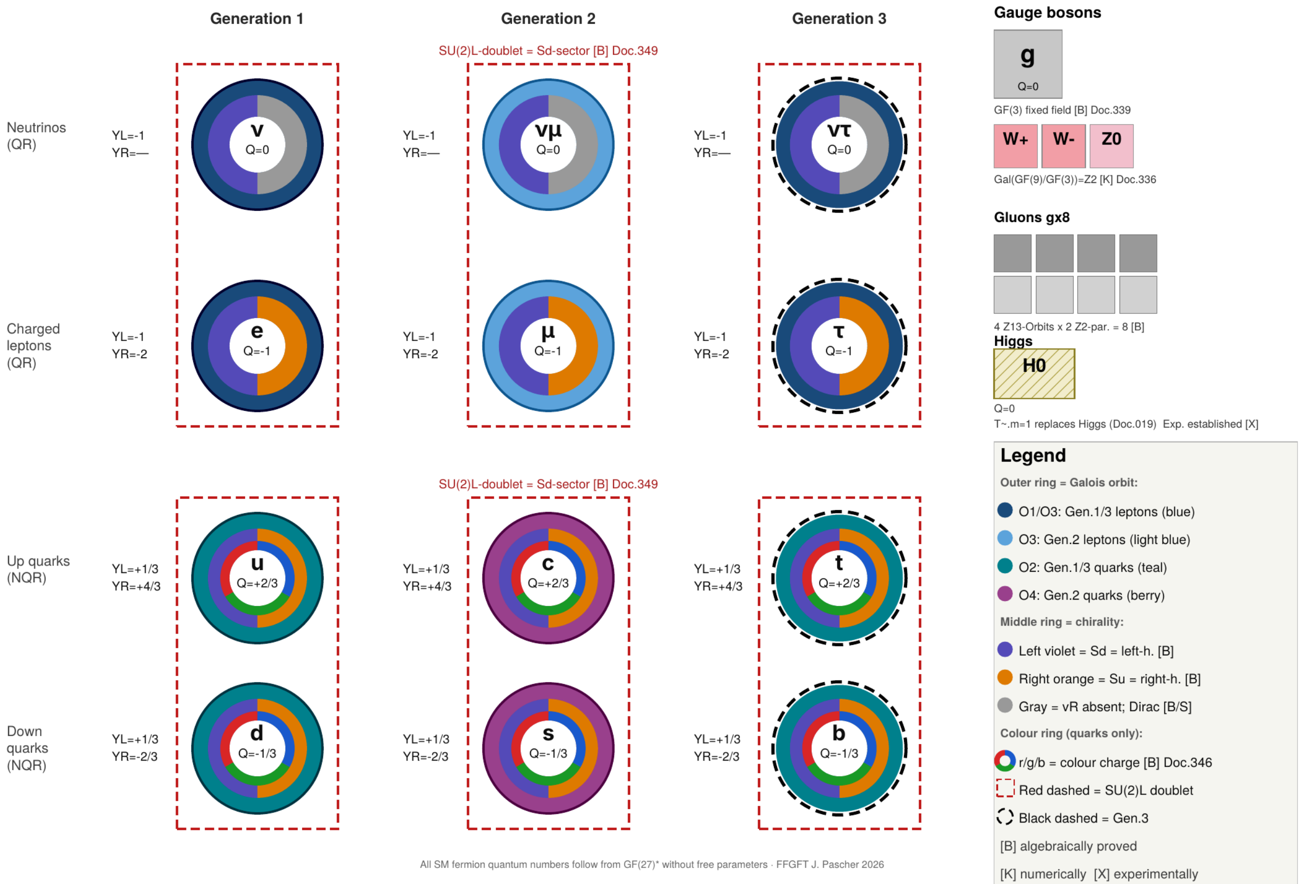
CKM complex phase δ_{CP} [S] — $\approx$ 68° from $GF(27)^*$ not directly derivable ($\mathbb{Z}_3$ phases yield 120°). Usable as phase anchor, status like v = 246 GeV.

Appendix E — Particle Diagram (Doc. 356)

All Standard Model fermions from $GF(27)^*$ — all quantum numbers algebraically forced. [B Doc. 346–349] Full resolution: `jpascher/T0-Time-Mass-Duality` → `2/pdf/356_SM_Particles_GF27_En.pdf`

Standard Model Particles from GF(27)* — all quantum numbers algebraically forced [B]

FFGFT · J. Pascher 2026 · Doc. 346/347/348/349



Appendix F — Sources and Acknowledgements

Primary Sources

The complete FFGFT corpus (over 360 documents, as of September 2026) is freely accessible:

GitHub https://github.com/jpascher/T0-Time-Mass-Duality
Complete TeX sources, Python test scripts, all documents in De and En, changelogs and register entries.

Zenodo DOI 10.5281/zenodo.17390357
Archived snapshot of the corpus (September 2026), citable and permanently available.

External Comparison: GALG Framework

Douglas Matzke (Information Physics Institute, IPI) has developed the GALG framework — an independent algebraic description of physical structures via Clifford algebras G(n). His real-time verification in GALG G(3) and G(6) provided the first external algebraic cross-check of the FFGFT Galois layer: the impossibility of order-26 elements in G(3) is algebraically forced and independent of FFGFT.

The results are documented in Doc. 341 (GF(27) in GALG and FFGFT, 1 September 2026) and in Doc. 326 (comparison FFGFT with hyperbit model).

Website and documents by Douglas Matzke:

Website https://www.quantumdoug.com
Central collection of his work on geometric algebra, hyperbits, limits of computation and quantum computing.

Dissertation D. Matzke, *Quantum Computation Using Geometric Algebra*, University of Texas at Dallas, 2002.
https://www.matzkefamily.net/doug/papers/PHD/phd_matzke_final.pdf

GALG framework D. Matzke, *$G(6, \mathbb{Z}_3\mathbb{C})$ and GALG framework*, IPI Letters 2022ff.

ANPA 2026 D. Matzke, *How the Vacuum Builds Space: Nilpotents, Quaternions, and the Planck Voxel*, ANPA Conference, 13 July 2026.

IPI 2026 D. Matzke, *Black Holes from Hyperbits*, IPI Annual Meeting 2026.

IPI talk Recording and summary:
https://informationphysicsinstitute.blogspot.com/2025/07/ipi-talk-doug-matzke-wwwquantumdougcom.html

Correspondence IPI mails of 1 and 3 September 2026 (GF(27) in GALG), documented in Doc. 341.

Information Physics Institute (IPI): https://www.ipigroup.net

Note on the Use of AI Assistance

Several documents of this corpus were created with assistance from Claude (Anthropic, Sonnet 4.6). The use follows a clear principle:

What the AI does: Writing and executing Python test scripts; algebraic calculations over large candidate spaces (thousands of blade searches, order determinations up to $\sim 10^{30}$); elaborating the LaTeXsources from results and author corrections; maintaining the file structure, changelogs and cross-references; preparing correspondence drafts that Johann Pascher reviews and approves.

What the author does: Poses all research questions; proofreads all results; draws all physical conclusions; decides what is calculated and written; evaluates every result; corrects errors; determines what the results mean physically. All statements carry explicit evidence markers [B], [K], [S] or [X] — every algebraic statement is secured by a Python test script with explicit assertions. If an assertion fails, the statement is not used.

The tool calculates and writes. The author decides.